

LARGE ERDŐS–STRAUS WITNESSES II: THE MULTIPLIER MODULUS IS INFINITELY OFTEN $\exp((\log p)^{1/4-o(1)})$

ANONYMOUS

ABSTRACT. For a prime p let $W(p)$ be the least modulus $M \equiv 3 \pmod{4}$ of a “multiplier” congruence that yields a solution of $4/p = 1/x + 1/y + 1/z$. We prove $W(p) \geq \exp(c(\log p)^{1/4}(\log \log p)^{-1/4})$ for infinitely many primes p in Mordell’s hard classes, assuming Gallagher’s prime number theorem for the family of all characters and the Nair–Tenenbaum bound for short sums of multiplicative functions of polynomials; more precisely, the least hard prime with $W(p) > T$ is at most $\exp(O((\log T)^4 \log \log T))$. The previous version had exponent $1/5$ and needed four results of Elsholtz and Tao. We also determine the Haar side: the profinite density $\delta^*(T)$ of the integers with $W(n) > T$ satisfies $(\log T)^3 / \log \log T \ll \log(1/\delta^*(T)) \ll (\log T)^3 (\log \log T)^5$. The upper bound comes from quarantining n to a *random square class* modulo a growing modulus: the residues $-4D \pmod{M}$ of the multiplier congruences are Jacobi non-residues, so no congruence is ever forced, and in expectation the residual masses are Haar masses. A local lemma with weights $\beta^{|\text{supp } E|}$ and a supermartingale bookkeeping control the cost. The lower bound comes from a Janson-type inequality for events on product spaces of one-hot variables, where Harris’ inequality fails. The transfer to primes uses, as before, the one-sided ℓ^2 sandwich of Bazzi and Razborov, an energy bound for avoidance indicators of cylinder events, and a linear transfer theorem based on Gallagher’s theorem. Finally we show that $1/4$ is the limit of this architecture: on every fibre, no minorant of $\mathbf{1}[W(n) > T]$ of level $\exp(c(\log T)^4 / \log \log T)$ has positive Haar mean. The proof is a planting argument, the dual of a lower-bound sieve of dimension $\asymp (\log T)^3$.

1. INTRODUCTION

The Erdős–Straus conjecture asserts that $4/n = 1/x + 1/y + 1/z$ is solvable in positive integers for every $n \geq 2$. It suffices to treat primes. It has been verified for $n \leq 10^{18}$ ([29]; [37] for 10^{17}), and Mordell’s identities settle every prime outside the six classes $p \equiv 1^2, 11^2, 13^2, 17^2, 19^2, 23^2 \pmod{840}$ [28] (cf. [32]). We call the primes in these classes *hard*. For the exceptional set see Vaughan [41] and [33]; for the average number of solutions see [11].

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For $M \equiv 3 \pmod{4}$ put $A_M = (M+1)/4$. If $uvw = A_M$ and $n \equiv -uv^{-1} \pmod{M}$, then $s = (nv+u)/M$ is a positive integer and

$$\frac{4}{n} = \frac{1}{suw} + \frac{1}{nsuw} + \frac{1}{nuvw} \quad (1)$$

(direct verification; this is the classical multiplier family, cf. [41, 11]). Define

$$W(n) = \min\{M \equiv 3 \pmod{4} : n \equiv -uv^{-1} \pmod{M} \text{ for some } uvw = A_M\}, \quad (2)$$

with $W(n) = +\infty$ if no such M exists. Thus $W(p) \leq T$ provides an explicit representation with modulus at most T . Every prime $p \equiv 1 \pmod{\text{lcm}\{M \leq T\}}$ has $W(p) > T$ (Lemma 2.3), so Linnik's theorem gives $W(p) \gg \log p$ infinitely often. In [2] we proved that for every fixed k , $W(p) \geq (\log p)^k e^{-C_k \log \log p / \log \log \log p}$ for infinitely many hard primes. Here we prove a sub-exponential rate.

Theorem 1.1 (Proved modulo Theorems 5.2 and 3.1). *There is an absolute constant $c > 0$ such that for infinitely many hard primes p*

$$W(p) \geq \exp\left(c \frac{(\log p)^{1/4}}{(\log \log p)^{1/4}}\right).$$

More precisely, for every sufficiently large T there is a hard prime p with $p > T$, $W(p) > T$ and $\log p \leq C(\log T)^4 \log \log T$.

Theorem 5.2 is Gallagher's prime number theorem for the family of all primitive characters of conductor at most Q_G [14], quoted from [26, Thm. 28.19] together with the standard facts on exceptional zeros (Theorem 5.1). No hypothesis on Siegel zeros is needed. Theorem 3.1 is the Nair–Tenenbaum bound [30] for sums of multiplicative functions of several polynomials, quoted in the form of Henriot [16]; it is used once, to bound Haar-weighted divisor sums over the moduli $M = 4n - 1$ (Lemma 3.2). The Elsholtz–Tao estimates used in the previous version are no longer needed. All constants are effective, except that we have not checked whether the implied constants in [30] are. We make no numerical claim.

Let $\delta^*(T)$ be the Haar measure of $\{n \in \hat{\mathbb{Z}}^\times : W(n) > T\}$ relative to the class $n \equiv 1 \pmod{24}$ (§3).

Theorem 1.2 (Haar exponent 3). *For large T ,*

$$c \frac{(\log T)^3}{\log \log T} \leq \log \frac{1}{\delta^*(T)} \leq C(\log T)^3 (\log \log T)^5.$$

The upper bound is proved modulo Theorem 3.1 (Theorem 3.3); the lower bound is proved modulo the fundamental lemma of sieve theory (Theorem 4.4).

The exponent of $\log p$ in Theorem 1.1 is governed not by the Haar density but by the level of the minorant used to transfer it to primes, and this level cannot be lowered.

Theorem 1.3 (Proved modulo Theorems 5.1, 5.2 and the fundamental lemma). *There are absolute $c, T_0 > 0$ such that for $T \geq T_0$, every modulus Q with $\log Q \leq T^{0.05}$ and every unit r modulo Q : if B is a linear combination of functions $n \mapsto \phi(n \bmod qQ)$ with $\log q \leq c(\log T)^4 / \log \log T$, and $B(n) \leq \mathbf{1}[W(n) > T]$ for all units $n \equiv r \pmod{Q}$, then the Haar mean of B on this class is ≤ 0 .*

This is Theorem 10.6. Consequently (Corollary 10.7) every argument that certifies a hard prime with $W(p) > T$ through a minorant of positive Haar mean and a transfer theorem costing $\log x \gg \log(\text{level})$, as ours does, has $\log x \gg (\log T)^4 / \log \log T$: such arguments cannot give $W(p) \geq \exp((\log p)^{1/4+\delta})$, and Theorem 1.1 is optimal among them up to a factor $(\log \log p)^{1/2}$ in $\log W(p)$. It also shows that the energy bound used for the sandwich cannot be sharpened for this purpose (Corollary 10.8). If primes behaved like a set of Haar density $\delta^*(T)$, Theorem 1.2 would suggest $\log W(p) \approx (\log p)^{1/3}$ for the extremal primes; reaching this needs prime input that is not a bounded-level minorant (Remark 10.9).

Versions. The first version of this note had exponent $1/14$; it used the Thorner–Zaman theorem [39] through the transfer theorem of [2]. The second had exponent $1/7$: it replaced Thorner–Zaman by the linear transfer theorem (Theorem 6.1 here), and bounded the approximation error by Håstad’s switching lemma. The third had exponent $1/5$: it replaced the switching lemma by the energy bound (§7) and used a graded class-of-one quarantine, whose cost was bounded with four results of Elsholtz and Tao. The present version keeps the energy bound, the sandwich and the transfer theorem, and changes the quarantine: a random square class, a β -weighted local lemma and a supermartingale bookkeeping (§2) with Haar masses bounded by the Nair–Tenenbaum theorem (§3). The third version also had an unconditional bound $\log W(p) \geq (1/\log 2 - o(1)) \log \log p \cdot \log \log \log p$ modulo Gallagher’s theorem alone; we drop it, since Theorem 1.1 now needs only Gallagher’s theorem and the published Nair–Tenenbaum theorem. New are also the Haar lower bound (§4), which was cited from companion work in the third version, and the ceiling (§10).

The method. Fix T . We restrict n to a class $n \equiv r \pmod{Q}$, with $Q = 8 \prod_{\ell} \ell^{a_{\ell}}$ and r a quadratic residue modulo every prime of Q . Then $W(n) > T$ whenever n avoids a finite family of congruence *events* $n \equiv -4D \pmod{m}$, one for each surviving pair (M, D) with $M \leq T$, $D \mid A_M^2$, where m is the part of M not absorbed by Q . The key observation (Lemma 2.2) is that the Jacobi symbol $(\frac{-4D}{M})$ is always -1 ; so if every prime of M divides Q , the event is excluded, and no event is ever forced, whatever square class r we choose. For $r = 1$ this is the class-of-one quarantine of [2]; Mordell’s hard classes are the case $Q = 840$.

We build (Q, r) by a random process (Lemma 2.9). While the weighted mass $\tilde{w}_{\ell} = \sum_{E \ni \ell} \beta^{|\text{supp } E|} \mathbb{P}(E)$ at a prime $\ell \leq Y = \mathcal{L}^{O(1)}$ exceeds $\eta = \frac{3}{4} \log \beta$,

we reveal one more digit of n at ℓ , uniformly among the squares. The β -weighted local lemma (Lemma 2.8) needs exactly $\tilde{w}_\ell \leq \eta$ at every coordinate. With $\beta = 1 + 1/\log \mathcal{L}$ the threshold is $\asymp 1/\log \mathcal{L}$ at every prime and every level, instead of $\asymp \log \ell/\mathcal{L}$ in the previous version. The masses form supermartingales (a step at a new prime at most doubles the mean of an event's probability, and this is paid by a factor $2^{\omega_Y(M)}$), so by optional stopping the expected cost is a Haar-weighted divisor sum, $\ll \mathcal{L}^3(\log \mathcal{L})^5$ by the Nair–Tenenbaum theorem. Primes $\ell > Y$ are never quarantined; a second-moment potential for pairs of events shows that their masses are small with high probability. The class of one, used before, inflates the masses by $\gcd(M, 4D + 1)$, which costs a factor $\mathcal{L}$ and requires the Elsholtz–Tao estimates.

To find a *prime* in the avoiding set we need a function $B = \sum_i c_i \mathbf{1}[n \equiv b_i(d_i)]$ with $B \leq \mathbf{1}[\text{avoid}]$ pointwise, mean $\gg \delta$ (the Haar density of the avoiding set on the fibre), Haar ℓ^1 norm $\mathbb{E}|B|$ comparable to its mean, and small moduli d_i . The linear transfer theorem (Theorem 6.1) then gives a prime $p \equiv r \pmod{Q}$ with $\log p \ll \log Z$, where Z is Q times the largest d_i . It is Linnik's theorem with weights: the function $B(n)\mathbf{1}[n \equiv r \pmod{Q}]$ is expanded in characters, every coefficient is bounded by the Haar ℓ^1 norm of B , whatever the number of cells, and Gallagher's theorem bounds the contribution of all characters of conductor at most a power of Z at once. Since r is a square, every real character that factors through $(\mathbb{Z}/Q)^\times$ is 1 at r , so the exceptional-zero analysis is the same as for the class of one.

As in the previous versions, B is the one-sided ℓ^2 sandwich of Bazzi [3], in the form of Razborov [35] (Lemma 8.2). For events $E_1, \dots, E_m$ with indicators A_i and arbitrary functions u_j ,

$$B = 1 - \sum_i A_i \left(1 - \sum_{j < i} A_j u_j \right)^2 \leq \mathbf{1}[\text{no } E_i \text{ occurs}],$$

and, if each u_j depends only on the coordinates not fixed by E_j , $\mathbb{E}[F - B] \leq m^2 \sum_j \mathbb{P}(E_j) \mathbb{E}(F^{(j)} - u_j)^2$, where $F^{(j)}$ is the indicator that no earlier event occurs, conditioned on E_j . The u_j are truncations of the Efron–Stein decomposition of $F^{(j)}$, and the error is an Efron–Stein tail, which we bound by the energy bound of the previous version (Theorem 7.5): if F is the avoidance indicator of any family of cylinder events on a finite product probability space, and $\lambda_v \geq 1$ are weights with $\prod_{v \in \text{supp } E} \lambda_v \leq 2$ for every event E , then

$$\sum_U \left(\prod_{v \in U} \lambda_v \right) \|F^{=U}\|_2^2 \leq 1.$$

Applied to the base- ℓ digits of the residues, with weights adapted to the moduli (Theorem 7.9), it shows that truncating $F^{(j)}$ to moduli at most e^τ costs $2^{-\tau/(2 \log T)}$. So moduli of size $\log d_i \ll \mathcal{L} \cdot (S + \mathcal{L})$ suffice, where $S \ll \mathcal{L}^3 \log \mathcal{L}$ is the residual mass. Altogether $\log Z \ll \mathcal{L}^4 \log \mathcal{L}$. The ceiling (Theorem 10.6) shows that the level $\mathcal{L}S$ is forced: conditioned on the small

primes, the events with one large prime factor form a sieve of dimension $\asymp \mathcal{L}^3 / \log \mathcal{L}$ on the primes above $T^{0.6}$, and a planting lemma (Lemma 10.1), the dual of a lower-bound sieve, shows that no function of lower level can be a minorant with positive mean.

Relation to the literature. *The energy bound.* The nearest prior art known to us is Lecomte and Tan [23], who bound the Fourier coefficients of a DNF on $\{\pm 1\}^n$ by the probability that a set S is covered by the terms satisfied at a random point; this is the device of our Lemma 7.2. Their cover counts are unsigned, and their degree concentration still comes from Håstad’s switching lemma, with an unspecified constant. In Lemma 7.2 the cover counts are signed, so that they vanish when a satisfied event misses the set; the polarization identity (Lemma 7.3) and the deletion–contraction bound (Lemma 7.4) then give the constant 1, on any finite product probability space. In Boolean language (Remark 7.8): every width- k DNF $g : \{\pm 1\}^n \rightarrow \{\pm 1\}$ has $\sum_{|S| > t} \hat{g}(S)^2 \leq 4 \cdot 2^{-(t+1)/k}$, under any product measure, while the switching-lemma route gives $2 \cdot 2^{-t/(20k)}$ [31, §4.4]. The base $2^{-1/k}$ is sharp over general product spaces (Remark 7.7). Theorem 7.5 and its corollaries are new to us, but our literature search was partial (we checked [23], [31] and some lecture notes; we did not search the literature on switching-lemma variants, Fourier growth or hypercontractivity with $\rho > 1$ systematically), and we make no priority claim.

The local lemmas. Lemma 2.8 is the asymmetric local lemma [1, Lemma 5.1.1] with the choice $x_E = \beta^{|\text{supp } E|} \mathbb{P}(E)$; the conditional form for events outside the family is standard (cf. [17]). Theorem 4.3 combines the lopsided local lemma of Erdős and Spencer [10] with the proof scheme of Janson’s inequality [22], which needs increasing events and Harris’ inequality; for atomic events on one-hot coordinates we use instead the correlation inequality of Lemma 4.1. We did not find this form in the literature, but did not search systematically. The planting lemma (Lemma 10.1) is the linear-programming dual of lower-bound sieves; constructions of k -wise independent laws with extremal all-zero or all-one probabilities of the same kind appear in [4, 34], and we claim no novelty for it. Its use to bound the level of minorants of a concrete arithmetic indicator, for every fibre, is new to us.

The rest. The sandwich (Lemma 8.2) is Razborov’s identity. To our knowledge tools from the theory of bounded independence (“polylogarithmic independence fools DNF and AC^0 ” [3, 35, 7]) have not been used before to construct sieve minorants for primes. Gallagher’s theorem is classically the engine of Linnik’s theorem on the least prime in a single progression. Here it serves as a transfer device for an arbitrary signed combination of progressions; the cost is governed by the Haar ℓ^1 norm of the minorant and its largest modulus, not by the number of cells. The nearest classical analogues are bounds for the least prime in a union of residue classes, or in a Chebotarev class of an abelian extension (Lagarias–Montgomery–Odlyzko [27]; Thorner–Zaman [40]); these concern nonnegative indicators. We have not searched

systematically for an earlier use of Gallagher’s theorem in this role. Earlier uses of the switching lemma in number theory known to us run in the opposite direction: they show that the Möbius function is uncorrelated with bounded-depth circuits [15, 6]. The Haar side, a local-lemma lower bound for the density of the uncovered set of a congruence system, has relatives in the work on covering systems: Filaseta, Ford, Konyagin, Pomerance and Yu [12], who to our knowledge already use the local lemma for this purpose, Hough [19] and Balister, Bollobás, Morris, Sahasrabudhe and Tiba [5]. We do not use those results. The Jacobi-symbol property of the multiplier classes (Lemma 2.2) is implicit in the classical fact that the identities cover all non-residue classes; its use to quarantine to random square classes is ours. The divisor-sum bounds use Nair–Tenenbaum [30, 16] in a standard way.

Status conventions. Every theorem header carries a label. *Proved* means proved in full here. *Proved modulo X* means proved here assuming the cited statement X, whose statement we checked in the source but whose proof we did not check. *Cited* marks external statements; *Classical* marks standard results for which we include a proof or a precise textbook reference. The cited inputs are Gallagher’s theorem (Theorem 5.2, used together with the classical Landau–Page facts of Theorem 5.1), the Nair–Tenenbaum theorem (Theorem 3.1) and, for Theorems 4.4 and 10.6, the fundamental lemma of sieve theory. In the labels, “modulo Theorem 5.2” includes Theorem 5.1. Classical results whose proof we include, or which are standard textbook facts (the local lemma, the Efron–Stein decomposition, Mertens’ theorem, the divisor bound, the Rosser–Schoenfeld bound for $\vartheta(x)$, mean values of non-negative multiplicative functions via Euler products), are not listed in the labels. *Proved implication* marks a statement about a class of arguments. *Assessment* marks heuristic comparisons, which are not claims. Nothing here bears on the truth of the Erdős–Straus conjecture: a prime with large $W(p)$ may well have other representations.

Notation. T is large and $\mathcal{L} = \log T$; $\log_2$ always denotes the binary logarithm (iterated logarithms are written out as $\log \log$). τ is the divisor function (and, in §§8–9, also a truncation parameter; no confusion arises). $\omega(m)$ and $\Omega(m)$ count prime factors without and with multiplicity; $\omega_Y(m)$ counts the prime factors $\leq Y$, and m_Y is the Y -smooth part of m . A y -rough integer has all prime factors $> y$. For $\lambda = (\lambda_v)$ and a finite set U we write $\lambda^U = \prod_{v \in U} \lambda_v$.

2. ATOMS, SQUARE CLASSES AND THE LOCAL LEMMA

For $M \equiv 3 \pmod{4}$ put $\mathcal{R}(M) = \{-4D \bmod M : D \mid A_M^2\}$. Note that $\gcd(A_M, M) = 1$ (as $4A_M = M + 1$), so every element of $\mathcal{R}(M)$ is a unit class.

Lemma 2.1 (Proved; cf. [2]). $\{-uv^{-1} \bmod M : uvw = A_M\} = \mathcal{R}(M)$. Consequently, for every integer n , $W(n) > T$ if and only if $n \bmod M \notin \mathcal{R}(M)$ for every $M \leq T$ with $M \equiv 3 \pmod{4}$.

Proof. Since $4A_M \equiv 1 \pmod{M}$, from $uvw = A_M$ we get $v^{-1} \equiv 4uw$, hence $-uv^{-1} \equiv -4u^2w$ and $D = u^2w$ divides A_M^2 . Conversely let $D \mid A_M^2$. For each prime q with $q^a \parallel A_M$ and $q^d \parallel D$ ($d \leq 2a$) put $u_q = \max(0, d-a)$, $w_q = d-2u_q$, $v_q = a-u_q-w_q$; these are non-negative, so $u = \prod q^{u_q}$, $v = \prod q^{v_q}$, $w = \prod q^{w_q}$ satisfy $uvw = A_M$ and $u^2w = D$. $\square$

An *atom* is a pair (M, D) with $M \leq T$, $M \equiv 3 \pmod{4}$ and $D \mid A_M^2$. There are at most T^2 atoms, since $\tau(A_M^2) \leq 2A_M \leq T$.

Lemma 2.2 (Event classes are Jacobi non-residues; proved). *Let (M, D) be an atom and let d be the squarefree part of D . Then*

- (a) *every prime $\ell \mid M$ satisfies $\ell \nmid 2D$ and $(\frac{-4D}{\ell}) = (\frac{-d}{\ell})$;*
- (b) *the Jacobi symbol $(\frac{-4D}{M})$ equals -1 .*

Consequently, if r is an integer coprime to M that is a quadratic residue modulo every prime dividing M , then $r \not\equiv -4D \pmod{M}$.

Proof. (a) $\gcd(M, 4A_M) = 1$ and every prime of D divides A_M , so $\ell \nmid 2D$; and $-4D = -d \cdot (2t)^2$ for an integer t . (b) By (a), $(\frac{-4D}{M}) = (\frac{-d}{M}) = (\frac{-1}{M})(\frac{d}{M}) = -(\frac{d}{M})$, as $M \equiv 3 \pmod{4}$. Every prime $p \mid d$ has $v_p(D)$ odd and $v_p(D) \leq 2v_p(A_M)$, so $p \mid A_M$ and $M = 4A_M - 1 \equiv -1 \pmod{p}$. For odd p , quadratic reciprocity, with $(M-1)/2$ odd, gives $(\frac{p}{M}) = (\frac{M}{p})(-1)^{(p-1)/2} = (\frac{-1}{p})(-1)^{(p-1)/2} = 1$. If $p = 2$, then $2 \mid A_M$, so $M \equiv 7 \pmod{8}$ and $(\frac{2}{M}) = 1$. Hence $(\frac{d}{M}) = 1$. For the consequence, $(\frac{r}{M}) = \prod_{\ell \mid M} (\frac{r}{\ell})^{v_\ell(M)} = 1 \neq (\frac{-4D}{M})$. $\square$

We checked (a) and (b) by computer on all 2 401 032 atoms with $M \leq 2 \cdot 10^5$ (no failures; this check is not used). The lemma is the mechanism behind two classical facts: the identities (1) cover the non-residue classes, and Mordell's hard classes modulo 840 are exactly the six square classes. For $r = 1$ it contains Lemma 2.3 below.

Lemma 2.3 (Class of one; proved). $1 \notin \mathcal{R}(M)$ for every $M \equiv 3 \pmod{4}$.

Proof. Lemma 2.2 with $r = 1$. $\square$

Definition 2.4 (Square-class quarantine and residual system). For an odd prime $\ell \leq T$ let $f_\ell = \lfloor \mathcal{L} / \log \ell \rfloor \geq 1$ be the largest exponent with $\ell^{f_\ell} \leq T$. A *square-class quarantine* is a pair (Q, r) , where $Q = 8 \prod_{\ell} \ell^{a_\ell}$ with integers $0 \leq a_\ell \leq f_\ell$ for the odd primes $\ell \leq T$, and r is a unit modulo Q with $r \equiv 1 \pmod{8}$ that is a quadratic residue modulo every odd prime dividing Q . The *coordinate primes* are the odd $\ell \leq T$ with $a_\ell < f_\ell$; write $\mathcal{P}$ for their set. For $\ell \in \mathcal{P}$ let

$$\Omega_\ell = \{x \in (\mathbb{Z}/\ell^{f_\ell})^\times : x \equiv r \pmod{\ell^{a_\ell}}\}$$

(all units if $a_\ell = 0$), with the uniform measure, and let $\mathbb{P}, \mathbb{E}$ be the product measure on $\Omega = \prod_{\ell \in \mathcal{P}} \Omega_\ell$, with coordinates X_ℓ . An atom *survives* (Q, r) if $-4D \equiv r \pmod{\gcd(M, Q)}$ and $M \nmid Q$. Its *support* is $\text{supp}(M, D) = \{\ell :$

$v_\ell(M) > a_\ell\}$, a nonempty subset of $\mathcal{P}$, with $s(M, D) = |\text{supp}(M, D)| \leq \omega(M)$; its *modulus* is $\prod_{\ell \in \text{supp}} \ell^{v_\ell(M)}$, and its *event* is

$$E_{M,D} = \{X_\ell \equiv -4D \pmod{\ell^{v_\ell(M)}} \text{ for all } \ell \in \text{supp}(M, D)\}.$$

For $\beta \geq 1$ the *weighted fibre mass* at $\ell \in \mathcal{P}$ and the *weighted total mass* are

$$\tilde{w}_\ell = \sum_{\substack{(M,D) \text{ surviving} \\ \ell \in \text{supp}(M,D)}} \beta^{s(M,D)} \mathbb{P}(E_{M,D}), \quad S_\beta = \sum_{\substack{(M,D) \\ \text{surviving}}} \beta^{s(M,D)} \mathbb{P}(E_{M,D}),$$

where surviving atoms are counted individually (two atoms may give the same event). For an integer n we put $X_\ell(n) = n \bmod \ell^{f_\ell}$, and $F(n) = 1$ if n lies in no event of a surviving atom, $F(n) = 0$ otherwise; on Ω , F is the indicator that no event $E_{M,D}$ occurs.

Every event is nonempty: $-4D$ is a unit modulo ℓ , and $-4D \equiv r \pmod{\ell^{a_\ell}}$ by survival. The modulus is at most $M \leq T$. The class-of-one quarantines of [2] and of the previous versions are the case $r = 1$.

Lemma 2.5 (Reduction and weights; proved). *Let (Q, r) be a square-class quarantine.*

- (i) *If $n \equiv r \pmod{Q}$ and $F(n) = 1$, then $W(n) > T$.*
- (ii) *For every surviving atom,*

$$\mathbb{P}(E_{M,D}) = \frac{\gcd(M, Q)}{M} \prod_{\ell \in \text{supp}(M,D), a_\ell=0} \frac{\ell}{\ell-1}.$$

- (iii) *If $105 \mid Q$, then $r \bmod 840$ is one of Mordell's six hard classes, so every prime $p \equiv r \pmod{Q}$ is hard.*

Proof. (i) Let $M \leq T$, $M \equiv 3 \pmod{4}$, and suppose $n \equiv -4D \pmod{M}$ with $D \mid A_M^2$. Reducing modulo $\gcd(M, Q)$ gives $-4D \equiv r$. If $M \mid Q$, then $r \equiv -4D \pmod{M}$, and every prime of M divides Q , so r is a quadratic residue modulo it; this contradicts Lemma 2.2. Otherwise the atom survives, and n lies in its event, so $F(n) = 0$. Conclude with Lemma 2.1. (ii) Let $\ell \in \text{supp}$ and $v = v_\ell(M)$. If $a_\ell = 0$, the class of $-4D$ modulo ℓ^v has probability $1/\varphi(\ell^v) = \ell^{-v} \cdot \ell/(\ell-1)$ in Ω_ℓ . If $a_\ell \geq 1$, then $|\Omega_\ell| = \ell^{f_\ell - a_\ell}$, and the class (which lies in Ω_ℓ by survival) has $\ell^{f_\ell - v}$ elements, so probability $\ell^{a_\ell - v}$. As M is odd, $\gcd(M, Q) = \prod_{\ell \mid M} \ell^{\min(v_\ell(M), a_\ell)}$, and $\min(v_\ell(M), a_\ell) = v_\ell(M)$ for $\ell \notin \text{supp}$. (iii) A unit modulo 840 that is $\equiv 1 \pmod{8}$ and a square modulo 3, 5 and 7 is a square modulo 840, and there are $1 \cdot 1 \cdot 2 \cdot 3 = 6$ such classes; they are the six listed classes. $\square$

Remark 2.6 (The graded class-of-one quarantine). In the previous version $r = 1$, and the exponents a_ℓ were raised while the fibre mass exceeded $c(a_\ell + 1) \log \ell / \mathcal{L}$. With $r = 1$ a surviving atom has $\gcd(M, Q) \mid g = \gcd(M, 4D + 1)$, so by (ii) its weight is up to g/M instead of the Haar weight $1/\varphi(M)$. Summed over the atoms, the Haar weights give $O(\mathcal{L}^3 \log \mathcal{L})$ (Lemma 3.2), while for the inflated weights we could only prove $O(\mathcal{L}^4)$, and

that needed the Elsholtz–Tao divisor-sum estimates [11]. The threshold $\propto \log \ell / \mathcal{L}$ cost a further factor $\mathcal{L}$ in the quarantine. A *random* square class removes the inflation in expectation (Lemma 2.9), Lemma 2.2 guarantees that no event is ever forced, and the β -weighted local lemma (Lemma 2.8) replaces the log-weighted threshold by a constant one. The graded class-of-one quarantine is therefore superseded, and we omit it.

The local lemma.

Lemma 2.7 (Local lemma with conditioning; classical). *Let $\mathcal{A}$ be a finite family of events in a product space, each depending on a finite set of coordinates; write $A' \sim A$ if they share a coordinate. Suppose $x_A \in [0, 1]$ satisfy $\mathbb{P}(A) \leq x_A \prod_{A' \sim A, A' \neq A} (1 - x_{A'})$. Then $\mathbb{P}(\bigcap_{A \in \mathcal{A}} \bar{A}) \geq \prod_{A \in \mathcal{A}} (1 - x_A)$, and $\mathbb{P}(A \mid \bigcap_{A' \in \mathcal{S}} \bar{A}') \leq x_A$ for every $\mathcal{S} \subseteq \mathcal{A}$ with $A \notin \mathcal{S}$. Moreover, for every event B depending on a set J of coordinates and every $\mathcal{S} \subseteq \mathcal{A}$, $\mathbb{P}(B \mid \bigcap_{A \in \mathcal{S}} \bar{A}) \leq \mathbb{P}(B) \prod_{A \in \Gamma(B)} (1 - x_A)^{-1}$, where $\Gamma(B)$ is the set of $A \in \mathcal{S}$ depending on some coordinate in J .*

Proof. The first two statements are [1, Lemma 5.1.1] and its proof. For the third (cf. [17, Thm. 2.1]), B is independent of $\bigcap_{A \in \mathcal{S} \setminus \Gamma(B)} \bar{A}$, so

$$\mathbb{P}\left(B \mid \bigcap_{A \in \mathcal{S}} \bar{A}\right) \leq \frac{\mathbb{P}(B) \mathbb{P}(\bigcap_{A \in \mathcal{S} \setminus \Gamma(B)} \bar{A})}{\mathbb{P}(\bigcap_{A \in \mathcal{S}} \bar{A})} = \frac{\mathbb{P}(B)}{\mathbb{P}(\bigcap_{A \in \Gamma(B)} \bar{A} \mid \bigcap_{A \in \mathcal{S} \setminus \Gamma(B)} \bar{A})},$$

and the chain rule with the second statement bounds the denominator below by $\prod_{A \in \Gamma(B)} (1 - x_A)$. $\square$

Lemma 2.8 (β -weighted local lemma; proved). *Let $\mathcal{A}$ be a finite family of events on a product probability space, each determined by the coordinates in a nonempty set $\text{supp } A$, and put $s(A) = |\text{supp } A|$. Let $1 < \beta \leq e^{1/3}$, $\eta = \frac{3}{4} \log \beta$, and $\tilde{w}_v = \sum_{A \ni v} \beta^{s(A)} \mathbb{P}(A)$, the sum over the A with $v \in \text{supp } A$. Suppose that $\tilde{w}_v \leq \eta$ for every coordinate v . Then*

$$\mathbb{P}\left(\bigcap_{A \in \mathcal{A}} \bar{A}\right) \geq \exp\left(-\frac{4}{3} \sum_{A \in \mathcal{A}} \beta^{s(A)} \mathbb{P}(A)\right),$$

and for every event B determined by a coordinate set J and every $\mathcal{S} \subseteq \mathcal{A}$, $\mathbb{P}(B \mid \bigcap_{A \in \mathcal{S}} \bar{A}) \leq \beta^{|J|} \mathbb{P}(B)$.

Proof. Put $x_A = \beta^{s(A)} \mathbb{P}(A)$. Every A has a coordinate $v \in \text{supp } A$, so $x_A \leq \tilde{w}_v \leq \eta \leq 1/4$. For $0 \leq x \leq 1/4$, $-\log(1 - x) \leq x/(1 - x) \leq \frac{4}{3}x$. For any coordinate set J , each A' sharing a coordinate with J is counted in some $\tilde{w}_v$, $v \in J$, so

$$\prod_{A' \text{ meets } J} (1 - x_{A'}) \geq \exp\left(-\frac{4}{3} \sum_{v \in J} \tilde{w}_v\right) \geq \exp\left(-\frac{4}{3} |J| \eta\right) = \beta^{-|J|}. \quad (3)$$

With $J = \text{supp } A$ this gives $\mathbb{P}(A) = x_A \beta^{-s(A)} \leq x_A \prod_{A' \sim A} (1 - x_{A'})$, the hypothesis of Lemma 2.7. Its conclusions give $\mathbb{P}(\bigcap_{A \in \mathcal{A}} \bar{A}) \geq \prod_{A \in \mathcal{A}} (1 - x_A) \geq \exp(-\frac{4}{3} \sum_{A \in \mathcal{A}} x_A)$ and, with (3), the conditional bound. $\square$

The criterion is per coordinate, with a threshold that depends neither on the coordinate nor on the level of its quarantine; the price is the weight $\beta^{s(A)}$. We shall take $\beta = 1 + 1/\log \mathcal{L}$, so that $\beta^{\omega(M)}$ is $O(1)$ on average, and $\eta \asymp 1/\log \mathcal{L}$.

The square-class process. Fix a parameter $Y \geq 7$ and $\beta = 1 + 1/\log \mathcal{L}$, $\eta = \frac{3}{4} \log \beta$. The *square-class process* is the following random sequence of square-class quarantines (Q_i, r_i) . Start from $Q = 8$, $r = 1$. Then make three *forced* steps at $\ell = 3, 5, 7$ with $a = 0$, and afterwards, while some odd prime $\ell \leq Y$ with $\ell \in \mathcal{P}$ has $\tilde{w}_\ell > \eta$ (computed for the current quarantine), make a step at the least such ℓ . A *step* at (ℓ, a) , where $a = a_\ell < f_\ell$ is the current exponent, replaces a_ℓ by $a + 1$ and r by a class $r' \bmod Q\ell$ with $r' \equiv r \pmod{Q}$, chosen uniformly among the classes of the current fibre at ℓ that are squares modulo ℓ : if $a = 0$ this is uniform among the $(\ell - 1)/2$ nonzero squares modulo ℓ , and if $a \geq 1$ among the ℓ lifts of $r \bmod \ell^a$ to ℓ^{a+1} , all of which are squares. Each step raises some $a_\ell < f_\ell$ with $\ell \leq Y$, so the process stops after at most $\sum_{\ell \leq Y} f_\ell$ steps.

Every (Q_i, r_i) is a square-class quarantine. After the forced steps, $840 \mid Q$ and $r \equiv 1 \pmod{24}$, since 1 is the only nonzero square modulo 3. Primes $\ell > Y$ are never stepped. By Lemma 2.2, no atom ever has all its coordinates quarantined and matched: the final system contains no event of probability one.

Notation. At stage i let μ_i be the uniform probability on the units n with $n \equiv r_i \pmod{Q_i}$ (on $\hat{\mathbb{Z}}^\times$, or modulo any multiple of $\text{lcm}(Q_i, M)$), and for an atom $E = (M, D)$ put $p_i(E) = \mu_i(n \equiv -4D \pmod{M})$. So $p_i(E) = \mathbb{P}(E_{M,D})$ if the atom survives (Q_i, r_i) , and $p_i(E) = 0$ otherwise (if $M \mid Q_i$ the class is excluded by Lemma 2.2). Let $P_H(E) = 1/\varphi(M)$ be the Haar probability, $s_i(E)$ the support size at stage i , and $u_i(E) = \#\{\ell \leq Y : \ell \mid M, a_\ell(i) = 0\}$. Write $\omega_Y(M)$ for the number of primes $\ell \leq Y$ dividing M , and M_Y for the Y -smooth part of M .

Lemma 2.9 (Bookkeeping for the square-class process; proved). (a) *For*

every fixed $\phi(E) \geq 0$, $G_i = \sum_E p_i(E) 2^{u_i(E)} \phi(E)$ is a supermartingale, and $G_0 = \sum_E P_H(E) 2^{\omega_Y(M)} \phi(E)$.

(b) *With Q_{end} the final modulus,*

$$\mathbb{E}[\log Q_{\text{end}}] \leq \log 840 + \frac{1}{\eta} \sum_E P_H(E) 2^{\omega_Y(M)} \beta^{\omega(M)} \log M_Y.$$

(c) $\mathbb{E}[\sum_E \beta^{s_{\text{end}}(E)} p_{\text{end}}(E)] \leq S_H^\beta := \sum_E P_H(E) 2^{\omega_Y(M)} \beta^{\omega(M)}.$

(d) *For every prime $\ell > Y$, $\mathbb{E}[\tilde{w}_\ell(\text{end})^2] \leq B_2(\ell)$, where*

$$B_2(\ell) = \sum_{\substack{E, E' \\ \ell \mid M, \ell \mid M'}} \beta^{\omega(M) + \omega(M')} 2^{\omega_Y(M) + \omega_Y(M')} P_H(E) P_H(E') \\ \times \varphi\left(\frac{\gcd(M, M')}{\ell^{\min(v_\ell(M), v_\ell(M'))}}\right).$$

Hence $\mathbb{P}(\exists \ell > Y : \tilde{w}_\ell(\text{end}) > \eta) \leq \eta^{-2} \sum_{\ell > Y} B_2(\ell)$.

All sums are over atoms $E = (M, D)$, $E' = (M', D')$.

Proof. (a) Consider a step at (ℓ, a) . Atoms with $\ell \nmid M$, or with $v_\ell(M) \leq a$, are unaffected: their class modulo $\ell^{v_\ell(M)}$ is already determined by the fibre, and the step only refines it at higher levels. Let $v_\ell(M) > a$. If $a \geq 1$, the new class is uniform among the ℓ classes of the fibre at level $a+1$, so $\mathbb{E}[p_{i+1}(E) \mid \text{past}] = p_i(E)$, and u is unchanged. If $a = 0$, the conditional probability of the event becomes ℓ^{1-v} (with $v = v_\ell(M)$) if $r' \equiv -4D \pmod{\ell}$ and 0 otherwise; the class $r' \bmod \ell$ is uniform on the $(\ell-1)/2$ squares, so

$$\begin{aligned} \mathbb{E}[p_{i+1}(E) \mid \text{past}] &= p_i(E) \varphi(\ell^v) \ell^{1-v} \frac{2}{\ell-1} \mathbf{1}\left[\left(\frac{-4D}{\ell}\right) = 1\right] \\ &= \left(1 + \left(\frac{-d}{\ell}\right)\right) p_i(E) \leq 2p_i(E) \end{aligned}$$

by Lemma 2.2(a), while u drops by one if $\ell \leq Y$ (forced steps and all other steps have $\ell \leq Y$). So each term $p_i(E) 2^{u_i(E)} \phi(E)$ is a supermartingale. At the start $Q = 8$, $r = 1$ and M is odd, so $p_0(E) = P_H(E)$; and $u_0(E) = \omega_Y(M)$.

(b) For a pair (ℓ, a) with $\ell \leq Y$ let $\tau_{\ell,a}$ be the stage at which the (unforced) step at (ℓ, a) is taken, ∞ if it is never taken. At that stage $\tilde{w}_\ell > \eta$, so

$$\log \ell \cdot \mathbf{1}[\tau_{\ell,a} < \infty] \leq \frac{\log \ell}{\eta} G_{\tau_{\ell,a}}^{(\ell,a)} \mathbf{1}[\tau_{\ell,a} < \infty],$$

$$G_i^{(\ell,a)} = \sum_{E: v_\ell(M) \geq a+1} p_i(E) 2^{u_i(E)} \beta^{\omega(M)},$$

since at stage $\tau_{\ell,a}$ the atoms with $\ell \in \text{supp } E$ are those with $v_\ell(M) \geq a+1$, and $\beta^s p \leq \beta^\omega 2^u p$. The stopping times are bounded, and optional stopping for the nonnegative supermartingale $G^{(\ell,a)}$ (part (a) with $\phi = \beta^{\omega(M)} \mathbf{1}[v_\ell(M) \geq a+1]$) gives $\mathbb{E}[G_{\tau_{\ell,a}}^{(\ell,a)} \mathbf{1}[\tau < \infty]] \leq G_0^{(\ell,a)}$. Summing over $\ell \leq Y$ and $a < v_\ell(M)$, and using $\sum_{\ell \leq Y} v_\ell(M) \log \ell = \log M_Y$, gives the claim; the forced steps cost $\log 105$, and $\log Q \leq \log 8 + \log 105 + \sum_{\text{steps}} \log \ell$.

(c) Part (a) with $\phi = \beta^{\omega(M)}$, and $s_{\text{end}} \leq \omega$.

(d) The prime ℓ is never stepped, so for $\ell \mid M$, $p_i(E) = p_i(E^{(\ell)}) / \varphi(\ell^{v_\ell(M)})$, where $E^{(\ell)}$ is the event with its condition at ℓ removed, and

$$\tilde{w}_\ell^2 \leq \sum_{E, E'} \beta^{\omega(M) + \omega(M')} p(E) p(E'),$$

the sum over atoms with $\ell \mid M$, $\ell \mid M'$. For two events F, F' (conditions at sets of primes, with given residues) put $\Pi_i = p_i(F) p_i(F') R_i$, where $R_i = \prod_{\ell'} \rho_i(\ell')$ over the primes ℓ' constrained by both, and $\rho_i(\ell')$ is the number of classes modulo ℓ'^j in the current fibre, $j \leq \min(v_{\ell'}(F), v_{\ell'}(F'))$ being the depth to which the two prescribed residues agree ($\rho_i(\ell') = 1$ once $a_{\ell'} \geq j$). Then $p_i(F) p_i(F') \leq \Pi_i$, and $\Pi_i 2^{u_i(F) + u_i(F')}$ is a supermartingale. Indeed, consider a step revealing level $a+1$ at ℓ' . If neither event constrains ℓ' at this level, nothing changes. If only one does, its factor behaves as in

(a), and its ρ is already 1. If both do and they agree at this level, both probabilities are multiplied by the same factor $N \cdot \mathbf{1}[\text{match}]$, where N is the number of classes of the current fibre at this level ($N = \ell' - 1$ if $a = 0$, $N = \ell'$ if $a \geq 1$), and ρ is divided by N ; so $\Pi \mapsto N\mathbf{1}[\text{match}]\Pi$, whose mean is Π if $a \geq 1$ and $(1 + (\frac{-d}{\ell'}))\Pi \leq 2\Pi$ if $a = 0$, in which case both u 's drop. If both constrain ℓ' and disagree at this level, they cannot both match, so $\Pi \mapsto 0$. At the start $R_0 \leq \prod \varphi(\ell'^j) \leq \varphi(\gcd(M_F, M_{F'}))$, so $\mathbb{E}[\Pi_{\text{end}}] \leq 2^{\omega_Y(M_F) + \omega_Y(M_{F'})} P_H(F) P_H(F') \varphi(\gcd(M_F, M_{F'}))$. Apply this to $F = E^{(\ell)}$, $F' = E'^{(\ell)}$, multiply by $\varphi(\ell^{v_\ell(M)})^{-1} \varphi(\ell^{v_\ell(M')})^{-1}$ and sum. The last claim is Chebyshev's inequality and a union bound. $\square$

Part (d) is the reason for the random class: the second moment of the mass at a prime that is never quarantined is controlled by a Haar pair-correlation sum, a deterministic divisor sum in which $\gcd(M, 4D + 1)$ does not occur.

3. HAAR MASSES AND THE UPPER BOUND ON THE HAAR SIDE

The only analytic input of this section is the following theorem. For fixed $k \geq 1$, $A, B \geq 1$ and $\varepsilon > 0$, let $\mathcal{M}_k(A, B, \varepsilon)$ be the class of non-negative functions F of k variables with

$$F(a_1 b_1, \dots, a_k b_k) \leq \min(A^{\Omega(a_1 \cdots a_k)}, B(a_1 \cdots a_k)^\varepsilon) F(b_1, \dots, b_k)$$

whenever $\gcd(a_1 \cdots a_k, b_1 \cdots b_k) = 1$.

Theorem 3.1 (Nair–Tenenbaum [30, Thm. 1]; cited in the form of [16, (1.1)]). *Let $Q_1, \dots, Q_k \in \mathbb{Z}[X]$ be pairwise coprime irreducible polynomials, $Q = Q_1 \cdots Q_k$ of degree g , without fixed prime divisor. Let $\rho_{Q_j}(n)$ and $\rho(n)$ be the numbers of zeros of Q_j and of Q modulo n . Let $0 < \alpha, \delta < 1$, $A, B \geq 1$, $0 < \varepsilon \leq \alpha\delta/(12g^2)$ and $F \in \mathcal{M}_k(A, B, \varepsilon)$. Then, uniformly in $x \geq c_0 \|Q\|^\delta$ and $x^\alpha < y \leq x$,*

$$\begin{aligned} \sum_{x < n \leq x+y} F(|Q_1(n)|, \dots, |Q_k(n)|) &\ll y \prod_{p \leq x} \left(1 - \frac{\rho(p)}{p}\right) \\ &\times \sum_{n_1 \cdots n_k \leq x} F(n_1, \dots, n_k) \frac{\rho_{Q_1}(n_1) \cdots \rho_{Q_k}(n_k)}{n_1 \cdots n_k}, \end{aligned}$$

where c_0 and the implied constant depend only on $k, g, A, B, \varepsilon, \alpha, \delta$ and the discriminant of Q .

We use it with $k = 2$, $Q_1(n) = n$, $Q_2(n) = 4n - 1$: these are coprime and irreducible, $Q = 4n^2 - n$ has discriminant 1 and no fixed prime divisor ($Q(1) = 3$, $Q(2) = 14$), $\rho(2) = 1$, $\rho(p) = 2$ for odd p , $\rho_{Q_1} \equiv 1$ and $\rho_{Q_2}(m) = \mathbf{1}[m \text{ odd}]$. We take $y = x$, $\alpha = \delta = 1/2$ and $\varepsilon = 1/200$.

Throughout, $\beta = 1 + 1/\log \mathcal{L}$, $2 \leq Y \leq T$, $H(M) = \beta^{\omega(M)} 2^{\omega_Y(M)}$, and

$$w(M) = \tau(A_M^2) H(M) \frac{M}{\varphi(M)} \quad (M \equiv 3 \pmod{4}), \quad w(M) = 0 \text{ otherwise.}$$

Summing over $D \mid A_M^2$ gives $\sum_D H(M)P_H(E_{M,D}) = w(M)/M$. Sums over M run over $M \leq T$.

Lemma 3.2 (Haar masses; (A) proved modulo Theorem 3.1, (B) proved). (A)

We have $S_H^\beta = \sum_M w(M)/M \ll \mathcal{L}^3 \log Y$ and

$$\sum_E P_H(E) 2^{\omega_Y(M)} \beta^{\omega(M)} \log M_Y \ll \mathcal{L}^3 (\log Y)^4.$$

(B) $\sum_{\ell > Y} \text{prime} B_2(\ell) \leq \frac{\mathcal{L}+1}{Y} \Xi$, where $\Xi = \sum_M w(M)^2 \tau(M) \omega(M)/M \ll \mathcal{L}^{C_0}$ for an absolute constant C_0 , uniformly in $2 \leq Y \leq T$.

Proof. (A) Write $n = A_M$, so $M = 4n - 1$, and $f_2(m) = H(m)m/\varphi(m)$, a multiplicative function with $f_2(p^k) = \beta 2^{[p \leq Y]} p/(p-1) \leq 6$ for odd p (as $\beta \leq 2$). The function $F(n_1, n_2) = \tau(n_1^2) f_2(n_2)$ satisfies $F(a_1 b_1, a_2 b_2) = \tau(a_1^2) f_2(a_2) F(b_1, b_2)$ when $\gcd(a_1 a_2, b_1 b_2) = 1$, and $\tau(a_1^2) f_2(a_2) \leq 8^{\Omega(a_1 a_2)}$ and $\leq B(a_1 a_2)^{1/200}$ for an absolute B (divisor bound). So $F \in \mathcal{M}_2(8, B, 1/200)$, uniformly in T and Y . Theorem 3.1 on $x < n \leq 2x$ gives

$$\begin{aligned} \sum_{x < n \leq 2x} \tau(n^2) f_2(4n-1) &\ll \frac{x}{(\log x)^2} \sum_{n_1 \leq x} \frac{\tau(n_1^2)}{n_1} \sum_{n_2 \leq x} \frac{f_2(n_2)}{n_2} \\ &\ll \frac{x}{(\log x)^2} (\log x)^3 (\log x)^\beta (\log Y)^\beta, \end{aligned}$$

by the Euler products $\prod_{p \leq x} (1 + 3/p + \dots)$ and

$$\prod_{p \leq Y} \left(1 + \frac{2\beta}{p-1} + O(p^{-2})\right) \prod_{Y < p \leq x} \left(1 + \frac{\beta}{p-1} + O(p^{-2})\right).$$

Since $(\log x)^{\beta-1}, (\log Y)^{\beta-1} \leq \mathcal{L}^{1/\log \mathcal{L}} = e$, this is $\ll x(\log x)^2 \log Y$. As $\tau(A_M^2)H(M)/\varphi(M) = \tau(n^2)f_2(M)/M$ with $M \asymp x$, the block contributes $\ll \mathcal{L}^2 \log Y$; the $O(\mathcal{L})$ dyadic blocks (and the bounded range $x < c_0$) give the first bound. For the second, use $\log M_Y \leq \log Y \cdot \Omega(M_Y) \leq \log Y \cdot \tau(M_Y)$ (as $\sum k_i \leq \prod (k_i + 1) - 1$), and repeat the argument with $f_2(m)\tau(m_Y)$ in place of f_2 . Its values at p^k are at most $6(k+1) \leq 12^k$ and $\ll_\varepsilon p^{k\varepsilon}$, so the class condition holds uniformly; its Euler factor at $p \leq Y$ is $1 + 4\beta/(p-1) + O(p^{-2})$, so $\sum_{n_2 \leq x} f_2(n_2)\tau((n_2)_Y)/n_2 \ll (\log x)(\log Y)^3$, and the bound $\mathcal{L}^3 (\log Y)^4$ follows.

(B) For an atom pair with $\ell \mid M$, $\ell \mid M'$ write $M = \ell m$, $M' = \ell m'$. Then $\gcd(M, M')/\ell^{\min(v_\ell(M), v_\ell(M'))}$ divides $\gcd(m, m')$, so its φ is at most $\gcd(m, m') = \sum_{e \mid (m, m')} \varphi(e)$. Summing over D, D' ,

$$B_2(\ell) \leq \sum_e \varphi(e) V(e\ell)^2, \quad V(q) = \sum_{q \mid M} \frac{w(M)}{M} = \sum_{k \leq T/q} \frac{w(qk)}{qk}.$$

By Cauchy-Schwarz in k , $V(q)^2 \leq (\mathcal{L} + 1) \sum_k w(qk)^2 / (q^2 k)$. Put $N = \ell k$: the term $\varphi(e)w(N)^2 / (e^2 \ell^2 k)$ equals $(w(N)^2 / N) \varphi(e) / (e\ell) \leq (w(N)^2 / N) / \ell$,

and for fixed ℓ there are at most $\tau(N)$ factorisations $N = e\ell k$. So

$$\sum_{\ell > Y} B_2(\ell) \leq (\mathcal{L} + 1) \sum_N \frac{w(N)^2 \tau(N)}{N} \sum_{\ell | N, \ell > Y} \frac{1}{\ell} \leq \frac{\mathcal{L} + 1}{Y} \Xi.$$

For Ξ , use $\omega \leq \tau$, $2^{\omega_Y} \leq 2^\omega$ and Cauchy–Schwarz with $M = 4n - 1 \geq 3n$:

$$\Xi \leq \left(\sum_{n \leq T} \frac{\tau(n^2)^4}{n} \right)^{1/2} \left(\sum_{m \leq 4T} \frac{g(m)^2}{m} \right)^{1/2},$$

$$g(m) = \beta^{2\omega(m)} 4^{\omega(m)} \left(\frac{m}{\varphi(m)} \right)^2 \tau(m)^2.$$

Both are mean values of non-negative multiplicative functions f with $f(p^k) \leq C(k+1)^C$ for an absolute C ; bounding them by $\prod_{p \leq 4T} (1 + f(p)/p + f(p^2)/p^2 + \dots)$ gives $\ll \mathcal{L}^{C_0}$, with C_0 absolute (since $\beta \leq 2$) and independent of Y . $\square$

The proof of (A) is the only place where Theorem 3.1 is used. All masses here are Haar masses, so the Elsholtz–Tao estimates are not needed.

Let $\delta^*(T)$ be the Haar measure of $\{n \in \hat{\mathbb{Z}}^\times : W(n) > T\}$ relative to the class $n \equiv 1 \pmod{24}$, that is, the proportion, among the units modulo $L = \text{lcm}(24, M \leq T)$ that are $\equiv 1 \pmod{24}$, of those with $W(n) > T$ (the condition $W(n) > T$ depends only on $n \pmod{L}$). On all of $\hat{\mathbb{Z}}^\times$ the proportion is exactly half of this, because the atom $(3, 1)$ excludes $n \equiv 2 \pmod{3}$.

Theorem 3.3 (Haar side, upper bound; proved modulo Theorem 3.1). *For large T there is a square-class quarantine (Q, r) with $840 \mid Q$, $r \equiv 1 \pmod{24}$, all prime factors of Q at most $Y := \mathcal{L}^{C_0+4}$ (with C_0 from Lemma 3.2), and, with $\beta = 1 + 1/\log \mathcal{L}$ and $\eta = \frac{3}{4} \log \beta$,*

$$\tilde{w}_\ell \leq \eta \quad (\ell \in \mathcal{P}), \quad S_\beta \leq C \mathcal{L}^3 \log \mathcal{L}, \quad \log Q \leq C \mathcal{L}^3 (\log \mathcal{L})^5, \quad (4)$$

where C is absolute. Consequently $\delta = \mathbb{E}F \geq \exp(-\frac{4}{3}S_\beta)$ and

$$\log \frac{1}{\delta^*(T)} \leq \log \varphi(Q) + \frac{4}{3}S_\beta \ll \mathcal{L}^3 (\log \mathcal{L})^5.$$

Proof. Run the square-class process with this Y . As $1/\eta \asymp \log \mathcal{L}$ and $\log Y \asymp \log \mathcal{L}$, Lemma 2.9(b,c) and Lemma 3.2(A) give $\mathbb{E}[\log Q_{\text{end}}] \ll \mathcal{L}^3 (\log \mathcal{L})^5$ and $\mathbb{E}[S_\beta(\text{end})] \ll \mathcal{L}^3 \log \mathcal{L}$. By Lemma 2.9(d) and Lemma 3.2(B), $\mathbb{P}(\exists \ell > Y : \tilde{w}_\ell > \eta) \leq \eta^{-2} (\mathcal{L} + 1) \mathcal{L}^{C_0} / Y \leq 1/4$ for large T . By Markov's inequality each of the events $\log Q_{\text{end}} > 4\mathbb{E}[\log Q_{\text{end}}]$ and $S_\beta > 4\mathbb{E}[S_\beta]$ has probability at most $1/4$. So some realisation (Q, r) avoids all three. In it every coordinate $\ell \leq Y$ has $\tilde{w}_\ell \leq \eta$ because the process stopped, and every $\ell > Y$ by the choice of realisation; this is (4). Lemma 2.8 gives $\delta \geq \exp(-\frac{4}{3}S_\beta)$. By Lemma 2.5(i), the units $n \equiv r \pmod{Q}$ with $F(n) = 1$ have $W(n) > T$. Among the units $\equiv 1 \pmod{24}$ the class $r \pmod{Q}$ has relative Haar measure $\varphi(24)/\varphi(Q)$, as $r \equiv 1 \pmod{24}$, and within it the residues $n \pmod{\ell^{\ell_e}}$, $\ell \in \mathcal{P}$, are independent and uniform on Ω_ℓ . So $\delta^*(T) \geq (\varphi(24)/\varphi(Q)) \delta$. $\square$

Together with Theorem 4.4 below, this determines the Haar exponent, $\log \log(1/\delta^*(T)) = (3+o(1)) \log \mathcal{L}$. The quarantine costs $\mathcal{L}^3(\log \mathcal{L})^5$; the class-of-one route of the previous version gave $\log(1/\delta^*(T)) \ll \mathcal{L}^5 \log \mathcal{L}$ modulo the Elsholtz–Tao estimates.

4. A JANSON-TYPE INEQUALITY AND THE LOWER BOUND ON THE HAAR SIDE

In this section (X_v) are independent random variables with finite or countable ranges, and an *atomic event* is an event $E = \{X_v = a_{E,v} \text{ for } v \in S_E\}$ with S_E finite and single values $a_{E,v}$. For a family $\mathcal{F}$ put $\text{Av}(\mathcal{F}) = \bigcap_{E \in \mathcal{F}} \bar{E}$. Two atomic events *conflict* if they prescribe different values on some common coordinate (then $E \cap E' = \emptyset$), and *share a bit* if $S_E \cap S_{E'} \neq \emptyset$ and they agree on all common coordinates (then $\mathbb{P}(E \cap E') > 0$). Harris' inequality is not available here: for one-hot variables $\mathbb{P}(X = a \mid X \neq a') > \mathbb{P}(X = a)$. The following lemmas replace it.

Lemma 4.1 (Compatible events; proved). *Let A be atomic and let $\mathcal{C}$ be a family of atomic events none of which conflicts with A . Then $\mathbb{P}(A \cap \text{Av}(\mathcal{C})) \leq \mathbb{P}(A)\mathbb{P}(\text{Av}(\mathcal{C}))$.*

Proof. Condition on A . For $C \in \mathcal{C}$, $C \cap A = A \cap C'$, where C' is C with the coordinates in S_A deleted; C' is independent of $(X_v)_{v \in S_A}$ and $C' \supseteq C$. So $\mathbb{P}(\text{Av}(\mathcal{C}) \mid A) = \mathbb{P}(\bigcap \bar{C}') \leq \mathbb{P}(\bigcap \bar{C})$. $\square$

Lemma 4.2 (Lopsided local lemma; proved, standard). *Let $\mathcal{F}$ be a family of atomic events, $\Gamma(E)$ the set of $E' \in \mathcal{F}$ conflicting with E , and suppose $x_E \in (0, 1)$ satisfy $\mathbb{P}(E) \leq x_E \prod_{E' \in \Gamma(E)} (1 - x_{E'})$. Then for every atomic event A and every $\mathcal{S} \subseteq \mathcal{F}$,*

$$\mathbb{P}(A \mid \text{Av}(\mathcal{S})) \leq \mathbb{P}(A) \prod_{E' \in \mathcal{S} \text{ conflicting with } A} (1 - x_{E'})^{-1}.$$

Proof. By Lemma 4.1, $\mathbb{P}(E \mid \text{Av}(\mathcal{S}')) \leq \mathbb{P}(E)$ whenever no event of $\mathcal{S}'$ conflicts with E , so the lopsided local lemma of Erdős and Spencer [10] applies with the graph Γ ; its proof gives $\mathbb{P}(E \mid \text{Av}(\mathcal{S})) \leq x_E$ for $E \notin \mathcal{S}$. For A , let $\mathcal{S}_1$ be the events of $\mathcal{S}$ conflicting with A and $\mathcal{S}_2 = \mathcal{S} \setminus \mathcal{S}_1$. Then $\mathbb{P}(A \mid \text{Av}(\mathcal{S})) \leq \mathbb{P}(A \mid \text{Av}(\mathcal{S}_2)) / \mathbb{P}(\text{Av}(\mathcal{S}_1) \mid \text{Av}(\mathcal{S}_2))$; the numerator is at most $\mathbb{P}(A)$ by Lemma 4.1, and the chain rule bounds the denominator below by $\prod_{S_1} (1 - x_E)$. $\square$

Theorem 4.3 (Janson-type inequality for atomic events; proved). *Under the hypothesis of Lemma 4.2 put $K = \max_{E \in \mathcal{F}} \prod_{E' \in \Gamma(E)} (1 - x_{E'})^{-1}$, $\mu = \sum_{E \in \mathcal{F}} \mathbb{P}(E)$, and $\Delta = \sum \mathbb{P}(E \cap E')$ over the unordered pairs of distinct events of $\mathcal{F}$ that share a bit. Then*

$$-\log \mathbb{P}(\text{Av}(\mathcal{F})) \geq \mu - K\Delta, \quad -\log \mathbb{P}(\text{Av}(\mathcal{F})) \geq \min\left(\frac{\mu}{2}, \frac{\mu^2}{4K\Delta}\right).$$

Proof. Order $\mathcal{F} = \{E_1, \dots, E_N\}$ and put $B_i = \text{Av}(\{E_j : j < i\})$, so $-\log \mathbb{P}(\text{Av}(\mathcal{F})) = -\sum_i \log(1 - \mathbb{P}(E_i | B_i)) \geq \sum_i \mathbb{P}(E_i | B_i)$. Let $B_i^d = \text{Av}(\{E_j : j < i, S_j \cap S_i = \emptyset\}) \supseteq B_i$. Then $\mathbb{P}(E_i | B_i) \geq \mathbb{P}(E_i) \mathbb{P}(B_i | E_i) / \mathbb{P}(B_i^d)$. Given E_i , the events E_j ($j < i$) conflicting with E_i are impossible, those sharing a bit become E'_j (E_j with the coordinates of S_i deleted, so $\mathbb{P}(E_i) \mathbb{P}(E'_j) = \mathbb{P}(E_i \cap E_j)$), and B_i^d is independent of E_i . Hence

$$\mathbb{P}(B_i | E_i) \geq \mathbb{P}(B_i^d) - \sum_{j < i, j \text{ shares a bit with } i} \mathbb{P}(E'_j \cap B_i^d).$$

B_i^d is the avoidance event of a subfamily of $\mathcal{F}$, and every event conflicting with E'_j conflicts with E_j , so Lemma 4.2 gives $\mathbb{P}(E'_j | B_i^d) \leq K \mathbb{P}(E'_j)$. Therefore $\mathbb{P}(E_i | B_i) \geq \mathbb{P}(E_i) - K \sum_{j < i, j \sim i} \mathbb{P}(E_i \cap E_j)$, and summing over i gives the first bound. For the second, a subfamily $\mathcal{F}'$ satisfies the hypothesis with the same x_E and the same or smaller K , and $\mathbb{P}(\text{Av}(\mathcal{F})) \leq \mathbb{P}(\text{Av}(\mathcal{F}'))$. For a random $\mathcal{F}'$ with inclusion probability p , $\mathbb{E}\mu' = p\mu$ and $\mathbb{E}\Delta' = p^2\Delta$, so some $\mathcal{F}'$ has $\mu' - K\Delta' \geq p\mu - p^2K\Delta$; take $p = \min(1, \mu/(2K\Delta))$. $\square$

Δ counts only compatible pairs; conflicting pairs, which carry the structure “one coordinate, many residues”, enter only through K . If $x_E = 2\mathbb{P}(E)$, $\mathbb{P}(E) \leq 1/8$ and $\sum_{E' \in \Gamma(E)} \mathbb{P}(E') \leq 1/8$ for all E , the hypothesis holds and $K \leq e^{1/3}$. Theorem 4.3 was checked by exact enumeration on small random instances (not used). We did not find it in the literature in this form; the Janson inequality itself [22] needs positively correlated (increasing) events, and the lopsided local lemma alone gives upper bounds on $\mathbb{P}(E | \cdot)$, not lower bounds. We make no priority claim.

The lower bound. For the ES events the coordinates are $X_\ell = n \bmod \ell^f$ (f large) for odd primes ℓ , independent and uniform on units (on $X_3 \equiv 1 \pmod{3}$ in the normalisation of $\delta^*(T)$), and $E_{M,D} = \bigcap_{\ell|M} \{X_\ell \equiv -4D \pmod{\ell^{v_\ell(M)}}\}$. For squarefree M with $3 \nmid M$ these are atomic events in the variables $X_\ell \bmod \ell$, and $\mathbb{P}(E_{M,D}) = 1/\varphi(M)$. Distinct pairs $(M, -4D \bmod M)$ are distinct events.

Theorem 4.4 (Haar side, lower bound; proved modulo the fundamental lemma). *There are absolute constants $c > 0$ and T_0 such that for $T \geq T_0$*

$$\log \frac{1}{\delta^*(T)} \geq c \frac{\mathcal{L}^3}{\log \mathcal{L}}.$$

The only external input is the fundamental lemma of sieve theory [13, Lemma 6.3, Cor. 6.10] (cited), used for a lower bound for rough numbers in progressions to small moduli. T_0 is astronomically large.

The family. Put $y = \mathcal{L}^5$, $N_0 = \mathcal{L}^2$, and for an integer D write $D^* = \prod_p p^{\lfloor v_p(D)/2 \rfloor}$. Then $D | A^2 \iff D^* | A$, and each $n \geq 1$ equals D^* for exactly $2^{\omega(n)}$ integers D (write $D = \kappa t^2$ with κ squarefree; then $D^* = \kappa t$).

Let $\mathcal{F}$ be the set of events $E_{M,D}$ with

$$\sqrt{T} \leq M \leq T, \quad M \text{ squarefree with all prime factors } > y, \quad M \equiv 3 \pmod{4},$$

$$n := D^* \in [N_0, T^{1/10}], \quad n \mid A_M.$$

We use: (F1) $D \leq n^2 \leq T^{1/5} < M/4$, so different D give different residues $-4D \pmod{M}$, and $\mathbb{P}(E_{M,D}) = 1/\varphi(M)$; (F2) for y -rough squarefree $m \leq T$, $\omega(m) \leq \mathcal{L}/\log y$ and $\varphi(m) \geq m/2$; (F3) $\sum_{n \leq x} 2^{\omega(n)}/n \leq (1 + \log x)^2$ and $\sum_{n \leq x} 2^{\omega(n)} \leq x(1 + \log x)$; (F4) for a progression $v \equiv c \pmod{d}$ in $[v_0, V]$ with least element $v_1 \geq v_0$, $\sum 1/v \leq 1/v_1 + \log(V/v_0)/d$; (F5) if $\omega(m)y^{-1} \leq 1$, then $\sum_{g|m, g>1, g \text{ squarefree and } y\text{-rough}} g^{-1} \leq (1 + 1/y)^{\omega(m)} - 1 \leq 2\omega(m)/y$.

Lemma 4.5 (Mass; proved modulo the fundamental lemma). *We have $\mu = \sum_{E \in \mathcal{F}} \mathbb{P}(E) \geq c_1 \mathcal{L}^3 / \log y$.*

Proof. By (F1), $\mu \geq \sum_{N_0 \leq n \leq T^{1/10}} 2^{\omega(n)} \sum_M 1/M$, over y -rough squarefree $M \in [\sqrt{T}, T]$ with $M \equiv -1 \pmod{4n}$; this congruence implies $M \equiv 3 \pmod{4}$ and $n \mid A_M$. For a dyadic block $(X, 2X] \subseteq [\sqrt{T}, T]$, sift the progression (of length $X/(4n) \geq X^{4/5}/4$) by the primes $p < y$, $p \nmid 2n$, at level $X^{1/2}$; as $s = \log X/(2 \log y) \rightarrow \infty$, the fundamental lemma gives at least $c_2 X/(n \log y)$ y -rough elements. Those divisible by p^2 with $p \geq y$ number at most $X/(4ny) + \sqrt{2X}$, which is negligible as $n \leq X^{1/5}$. So each of the $\gg \mathcal{L}$ blocks contributes $\geq c_2/(4n \log y)$, and $\sum_{N_0 \leq n \leq T^{1/10}} 2^{\omega(n)}/n \gg \mathcal{L}^2$. $\square$

Lemma 4.6 (Local-lemma hypothesis; proved). *For $T \geq T_0$, every $E \in \mathcal{F}$ has $\mathbb{P}(E) \leq 1/8$ and $\sum_{E' \in \Gamma(E)} \mathbb{P}(E') \leq 1/8$. Hence Theorem 4.3 applies with $x_E = 2\mathbb{P}(E)$ and $K \leq e^{1/3}$.*

Proof. $\mathbb{P}(E) \leq 2/\sqrt{T}$. Put $w_q = \sum_{E' \in \mathcal{F}: q|M'} \mathbb{P}(E')$. Events conflicting with E share a prime $q \mid M$, so $\sum_{\Gamma(E)} \mathbb{P} \leq (\mathcal{L}/\log y) \max_{q>y} w_q$. Writing $M' = qv$ with $qv \equiv -1 \pmod{4n'}$ and $qv \geq \sqrt{T}$, (F2) and (F4) give $w_q \leq (2/q) \sum_{D'} (\mathcal{L}/(4n') + \min(1, q/\sqrt{T}))$. By (F3) the first part is at most $\mathcal{L}^3/(2q)$, and the second at most $4\mathcal{L}T^{-2/5}$. So $\sum_{\Gamma(E)} \mathbb{P} \leq (\mathcal{L}/\log y)(\mathcal{L}^3/(2y) + 4\mathcal{L}T^{-2/5}) = O(1/(\mathcal{L} \log \mathcal{L}))$. $\square$

Lemma 4.7 (Pair sum; proved). $\Delta = \sum_{\text{bit-sharing pairs}} \mathbb{P}(E \cap E') \leq C\mathcal{L}^2$.

Proof. Let $E = E_{M,D}$, $E' = E_{M',D'}$ be distinct and share a bit; put $g = \gcd(M, M') > 1$, $M = gv$, $M' = gv'$. Then g is y -rough, so $g > y$, and compatibility means $g \mid D - D'$. Also $M \neq M'$: otherwise $M \mid D - D'$ and $E = E'$ by (F1). By (F2), $\mathbb{P}(E \cap E') = 1/(\varphi(g)\varphi(v)\varphi(v')) \leq 8/(gvv')$. For fixed g and D the admissible v satisfy $gv \equiv -1 \pmod{4n}$ and $gv \geq \sqrt{T}$, so by (F4)

$$\sigma(g, n) := \sum_v \frac{1}{v} \leq \frac{\mathcal{L}}{4n} + \min\left(1, \frac{g}{\sqrt{T}}\right). \quad (5)$$

(a) $D = D'$. Then $v \neq v'$ and $v \equiv v' \equiv -g^{-1} \pmod{4n}$. For fixed (v, n) , g runs over a progression modulo $4n$ with $g > y$, so $\sum_g 1/g \leq 1/y + \mathcal{L}/(4n)$; and $\sum_{v < v', v' \equiv v \pmod{4n}} 1/(vv') \leq (1 + \mathcal{L})^2/(4n)$. So $\Delta_a \leq 8 \sum_{N_0 \leq n \leq T^{1/10}} 2^{\omega(n)} (1/y + \mathcal{L}/(4n)) (1 + \mathcal{L})^2/(4n) \ll \mathcal{L}^4/y + \mathcal{L}^3 \log N_0/N_0 \ll \mathcal{L} \log \mathcal{L}$.

(b) $D \neq D'$. Now $g \mid m = |D - D'|$ with $0 < m < T^{1/5}$, so $g < T^{1/5}$ and (5) gives $\sigma(g, n) \leq \mathcal{L}/(4n) + T^{-3/10}$. Summing over g with (F5),

$$\Delta_b \leq \frac{2\mathcal{L}}{y} \left[\frac{\mathcal{L}^2}{2} \left(\sum_D \frac{1}{n} \right)^2 + 4T^{-3/10} \mathcal{L} \# \mathcal{D} \sum_D \frac{1}{n} + 8T^{-3/5} (\# \mathcal{D})^2 \right],$$

where $\mathcal{D}$ is the set of D with $D^* \leq T^{1/10}$, so $\sum_D 1/n \leq (1 + \mathcal{L})^2$ and $\# \mathcal{D} \leq T^{1/10} (1 + \mathcal{L})$ by (F3). So $\Delta_b \ll \mathcal{L}^7/y + T^{-1/10} \mathcal{L}^5 \ll \mathcal{L}^2$. $\square$

Proof of Theorem 4.4. The events of $\mathcal{F}$ involve only primes $> y > 3$, so they are independent of $n \pmod{24}$, and $\delta^*(T) \leq \mathbb{P}(\text{Av}(\mathcal{F}))$. By Theorem 4.3 and Lemmas 4.5–4.7, $\log(1/\delta^*(T)) \geq \mu - e^{1/3} \Delta \geq c_1 \mathcal{L}^3/(5 \log \mathcal{L}) - C \mathcal{L}^2$. $\square$

Two restrictions make the pair sum small: $M \geq \sqrt{T} \geq (D^*)^5$ makes the first terms of the progressions in (5) harmless, and $D^* \geq \mathcal{L}^2$ removes the small- D hubs ($D = 1$ has residue -4 at every prime), whose same- D pairs would make $\Delta_a \asymp \mu$. The factor $\log \mathcal{L}$ is the price of y -roughness. A Monte Carlo estimate of $\delta^*(T)$ for $1023 \leq T \leq 32767$ fits $\log(1/\delta^*(T)) \approx 0.07 \mathcal{L}^3/\log \mathcal{L}$ (evidence only, not used; at these T it cannot distinguish $\mathcal{L}^3/\log \mathcal{L}$ from, say, $\mathcal{L}^{2.5}$).

5. THE ANALYTIC INPUT: GALLAGHER'S THEOREM

For a Dirichlet character χ put $\vartheta(x; \chi) = \sum_{p \leq x} \chi(p) \log p$, and let $E_0(\chi) = 1$ if χ is the trivial character modulo 1 and $E_0(\chi) = 0$ otherwise. $\sum_{\chi \bmod q}^*$ is a sum over primitive characters.

Theorem 5.1 (Landau–Page; classical, [8, Ch. 14], [25, Cor. 11.10]). *There are absolute effective positive constants c_1, c_2 such that, for every $Y \geq 3$, $\prod_{q \leq Y} \prod_{\chi \bmod q}^* L(s, \chi)$ has at most one zero in the region $\Re s > 1 - c_1/\log Y$, $|\Im s| \leq Y$. If this zero β_1 exists, it is real and simple, and its character χ_1 is quadratic. Moreover, every real zero β of $L(s, \chi)$, for χ a real primitive character of conductor $q \geq 3$, satisfies $1 - \beta \geq c_2^{-1} q^{-1/2} (\log q)^{-2}$.*

The first part is the classical zero-free region with the Landau–Page theorem. The height restriction matters: without it the statement would be a zero-free region of width $\asymp 1/\log Y$ uniformly in $|\Im s|$, which is not known. The “Exceptional Zero Statement” (28.61)–(28.62) of [26], which refers to [25, Cor. 11.10], omits the height restriction. The second part is the effective Page-type bound [8, Ch. 14], [26, (28.62)]. We use Theorem 5.1 only for *real* zeros: for the uniqueness of a real zero close to 1, for the quadraticity of its character, and for the lower bound for $1 - \beta$.

Theorem 5.2 (Gallagher [14]; cited in the form of [26, Thm. 28.19]). *There are absolute constants $c \geq 1$ and $\kappa_0 \geq 3$ with the following property. Let $\kappa \geq \kappa_0$ be fixed and $1 < Q_G^{6c} \leq x$. Then*

$$\sum_{q \leq Q_G} \sum_{\chi \bmod q}^* |\vartheta(x; \chi) - E_0(\chi)x| \ll_{\kappa} x \exp\left(-\frac{\log x}{\kappa \log Q_G}\right) + \left(\frac{\log x}{\log Q_G}\right)^2 \frac{x}{Q_G},$$

unless $\prod_{q \leq Q_G} \prod_{\chi \bmod q}^ L(s, \chi)$ has a real zero β_1 with $1 - \beta_1 < 1/(\kappa \log Q_G)$. In that case the term of the exceptional character χ_1 on the left is replaced by $|\vartheta(x; \chi_1) + x^{\beta_1}/\beta_1|$, and the right-hand side by*

$$(1 - \beta_1)(\log x) \left(x \exp\left(-\frac{\log x}{\log Q_G}\right) + \frac{x \log x}{Q_G \log Q_G} \right).$$

Source caveat. We quote [26], an unpublished draft (third volume of Montgomery–Vaughan), archived by us. We have not seen Gallagher’s paper [14], so we cite it without a theorem number. The printed proof of [26, Thm. 28.19] has visible slips. Its case split is at $1 - \beta_1 < 1/(3\kappa_0 \log Q_G)$, so it establishes the statement for $\kappa = 3\kappa_0$. The statement for every larger κ follows from that case. Let $\kappa \geq 3\kappa_0$, and suppose a real zero satisfies $1/(\kappa \log Q_G) \leq 1 - \beta_1 < 1/(3\kappa_0 \log Q_G)$. Then $x^{\beta_1}/\beta_1 \leq 2x \exp(-\log x/(\kappa \log Q_G))$, and $(1 - \beta_1)(\log x) x \exp(-\log x/\log Q_G) \ll x \exp(-\log x/(2 \log Q_G))$. So the exceptional form of the bound implies the non-exceptional one with parameter κ . We therefore fix

$$\kappa := \max(3\kappa_0, 1/c_1).$$

An exceptional zero as in Theorem 5.2 is then a real zero in the region of Theorem 5.1 with $Y = Q_G$, so it is unique and χ_1 is real. Write C_G for the implied constant in Theorem 5.2 with this κ .

Gallagher’s original statement may carry a range condition of the form $\exp(\sqrt{\log x}) \leq Q_G \leq x^b$; we have not checked this. Our use is robust to such a condition. In the proof of Theorem 6.1, $\log x / \log Q_G = \kappa L \geq 6c$, and $\log x \geq (\kappa L)^2$ by the hypothesis of that theorem, so $Q_G \geq \exp(\sqrt{\log x})$. The statement itself is the standard form of Gallagher’s prime number theorem for the whole family of characters of conductor $\leq Q_G$, which underlies the usual proofs of Linnik’s theorem.

6. A LINEAR TRANSFER THEOREM

Theorem 6.1 (Linear transfer; proved modulo Theorems 5.1 and 5.2). *Let $T \geq 1$ and $Q \geq 2$ with $8 \mid Q$, and let r be a unit modulo Q with $r \equiv 1 \pmod{8}$ that is a quadratic residue modulo every odd prime dividing Q . Let $B(n) = \sum_{i \in I} c_i \mathbf{1}[n \equiv b_i \pmod{d_i}]$ be a finite real combination of consistent unit cells: $\gcd(b_i, d_i) = 1$ and $b_i \equiv r \pmod{\gcd(d_i, Q)}$. Assume that $B(n) \leq \mathbf{1}[W(n) > T]$ for every $n \equiv r \pmod{Q}$ coprime to all d_i . Put $D = \text{lcm}_i d_i$, $N_0 = \text{lcm}(Q, D)$, $H = \{n \in (\mathbb{Z}/N_0)^\times : n \equiv r \pmod{Q}\}$, and let $\mathbb{E}_H$ be the mean over H . Put*

$$Z = Q \max_i d_i, \quad \mu = \mathbb{E}_H B, \quad A = \frac{\mathbb{E}_H |B|}{\mu},$$

and assume $\mu > 0$, $A \leq Z^{1/4}$, and the twist condition: for every real primitive character ψ of conductor $f > 1$ with $\gcd(f, Q) = 1$ and $f \mid d_i$ for some i ,

$$|\mathbb{E}_H[B\psi]| \leq \mu/4.$$

There is an absolute effective constant C_2 such that, if

$$\log x \geq C_2(1 + \log A) \log Z,$$

then some prime $p \leq x$ with $p \equiv r \pmod{Q}$ and $p \nmid D$ has $W(p) > T$.

In the second version $r = 1$ and the cells were coprime to Q ; then $\mathbb{E}_H$ is the mean over the units modulo D , and $\mu = \sum_i c_i / \varphi(d_i)$. Partial quarantines produce cells whose moduli share primes with Q ; consistency means that each cell meets the class $r \pmod{Q}$. The hypotheses on r are used only in (c) below: every real character that factors through $(\mathbb{Z}/Q)^\times$ takes the value 1 at r , exactly as at 1.

Neither the ℓ^1 mass $\sum_i |c_i| / \varphi(d_i)$ nor the number of cells enters; only the Haar quantity $A \geq 1$ does.

Proof. Parameters. Put $L = 2c + \log(400C_G(A+1))$ and $Q_G = x^{1/(\kappa L)}$. Since $A \geq 1$ we have $L \leq C_3(1 + \log A)$ for an absolute C_3 . If C_2 is large enough, the hypothesis gives the following, using $A \leq Z^{1/4}$ and $Z \geq 2$: $\log Q_G \geq (C_2/(\kappa C_3)) \log Z$, hence $Q_G \geq Z$ and $Q_G \geq 10^4 C_G(A+1)(\kappa L)^2$; $Q_G^{6c} \leq x$ (as $\kappa L \geq 6c$); $\log x \geq 16$; $\log x \geq (\kappa L)^2$ (as $(1 + \log A)^2 \leq (1 + \log A)(1 + \frac{1}{4} \log Z)$); and $x \geq C_4 A Z^4$, with C_4 as large as needed below.

Character expansion. The function $f(n) = B(n) \mathbf{1}[n \equiv r \pmod{Q}]$ on units modulo N_0 satisfies $f(n) = \sum_{\chi \bmod N_0} c(\chi) \chi(n)$. The reduction map $(\mathbb{Z}/N_0)^\times \rightarrow (\mathbb{Z}/Q)^\times$ is onto, and H is the fibre over r , so $|H| = \varphi(N_0)/\varphi(Q)$ and

$$c(\chi) = \frac{1}{\varphi(N_0)} \sum_{n \in H} B(n) \bar{\chi}(n) = \frac{\mathbb{E}_H[B\bar{\chi}]}{\varphi(Q)}.$$

Consequently:

- (a) $c(\chi) = 0$ unless the conductor of χ divides $\text{lcm}(d_i, Q)$ for some i . Indeed, by consistency and the Chinese remainder theorem, $\mathbf{1}[n \equiv b_i \pmod{d_i}] \mathbf{1}[n \equiv r \pmod{Q}]$ is the indicator of one unit class modulo $\text{lcm}(d_i, Q)$, and its χ -coefficient vanishes unless χ is trivial on $\{n \equiv 1 \pmod{\text{lcm}(d_i, Q)}\}$. So $c(\chi) \neq 0$ implies that the conductor of χ is at most $Qd_i \leq Z \leq Q_G$.
- (b) $|c(\chi)| \leq \mathbb{E}_H |B| / \varphi(Q) = A\mu / \varphi(Q)$, and $c(\chi_0) = \mu / \varphi(Q)$ for the principal character χ_0 .
- (c) Let χ be real with $c(\chi) \neq 0$, and let ψ^* be the real primitive character inducing it, of conductor f . Write $f = f_1 f_2$, where f_1 is composed of primes dividing Q and $\gcd(f_2, Q) = 1$, and $\psi^* = \psi_1 \psi_2$ accordingly. A real primitive conductor is $2^e f'$ with $e \in \{0, 2, 3\}$ and f' odd and squarefree; as $8 \mid Q$, $f_1 \mid Q$, and ψ_1 is a product of a character modulo 8 and Legendre symbols modulo odd primes of Q . On H we have $n \equiv r \pmod{Q}$, so $\psi_1(n) = \psi_1(r) = 1$, as $r \equiv 1 \pmod{8}$

and r is a quadratic residue modulo every odd prime of Q . So $\chi = \psi_2$ on H . If $f_2 = 1$, then $\chi = 1$ on H and $c(\chi) = \mu/\varphi(Q)$. If $f_2 > 1$, then $c(\chi) = \mathbb{E}_H[B\psi_2]/\varphi(Q)$. A cell with $f_2 \nmid d_i$ has $\mathbb{E}_H[\mathbf{1}[n \equiv b_i \pmod{d_i}]\psi_2] = 0$: there is a prime $p \mid f_2$ with $p \nmid d_i$; on H the residue of n modulo $p^{v_p(N_0)}$ is uniform among units and independent of n modulo $N_0/p^{v_p(N_0)}$, and the p -component of ψ_2 is a nontrivial character modulo p . So $f_2 \mid d_i$ for some i , $\gcd(f_2, Q) = 1$, and the twist condition gives $|c(\chi)| \leq \mu/(4\varphi(Q))$.

Let χ^* be the primitive character inducing χ . For a fixed modulus the map $\chi \mapsto \chi^*$ is injective, so by (a) at most $\sum_{r \leq Z} \varphi(r) \leq Z^2$ characters have $c(\chi) \neq 0$.

The prime sum. Let $S(x) = \sum_{p \leq x, p \nmid N_0} f(p) \log p = \sum_{\chi} c(\chi) \vartheta(x; \chi)$. Since $|\vartheta(x; \chi) - \vartheta(x; \chi^*)| \leq \log N_0 \leq \log Q + \log D \leq 2Z$ (as $D \mid \text{lcm}(1, \dots, \max d_i)$), so $\log D \leq \psi(\max d_i) < 1.04 \max d_i$ by Rosser–Schoenfeld [36],

$$S(x) = \frac{\mu x}{\varphi(Q)} + \sum_{\chi: c(\chi) \neq 0} c(\chi) (\vartheta(x; \chi^*) - E_0(\chi^*)x) + R_1, \quad (6)$$

where $|R_1| \leq Z^2 \cdot (A\mu/\varphi(Q)) \cdot 2Z = 2AZ^3\mu/\varphi(Q)$. Here $E_0(\chi^*) = 1$ only for $\chi = \chi_0$, and $|R_1| \leq \mu x/(400\varphi(Q))$ as $x \geq 800AZ^3$.

Case 0: no exceptional zero in the sense of Theorem 5.2. The χ^* in (6) are distinct primitive characters of conductor $\leq Q_G$. By (b) and Theorem 5.2, the middle sum is at most

$$\frac{A\mu}{\varphi(Q)} C_G x \left(e^{-L} + \frac{(\kappa L)^2}{Q_G} \right) \leq \frac{\mu x}{\varphi(Q)} \left(\frac{1}{400} + \frac{1}{10^4} \right) \leq \frac{\mu x}{200\varphi(Q)},$$

using $\log x / \log Q_G = \kappa L \geq L$. So $S(x) > 0$.

Exceptional case: a zero β_1 with character χ_1 and $1 - \beta_1 < 1/(\kappa \log Q_G)$. Put $u = (1 - \beta_1) \log x$; then $u < \log x / (\kappa \log Q_G) = L$, and $\beta_1 > 1/2$. The replaced right-hand side of Theorem 5.2 is $C_G u x (e^{-\kappa L} + \kappa L / Q_G)$. Since $u \leq \min(u, 1) L$ (recall $L \geq 2$), its product with $A\mu/\varphi(Q)$ is at most

$$\frac{\mu x}{\varphi(Q)} \min(u, 1) \cdot C_G A L \left(e^{-\kappa L} + \frac{\kappa L}{Q_G} \right) \leq \frac{\mu x}{200\varphi(Q)} \min(u, 1).$$

Here $C_G A L e^{-3L} \leq C_G A e^{-L} \leq 1/400$, and $C_G A \kappa L^2 / Q_G \leq 10^{-4}$. If no χ with $c(\chi) \neq 0$ has $\chi^* = \chi_1$, this bounds the middle sum of (6), and $S(x) > 0$ as in Case 0. Otherwise, by injectivity, exactly one such χ exists. It is real, because χ_1 is; hence $c(\chi)$ is real, as B and χ are. Let f_2 be as in (c). Its term is $c(\chi)(\vartheta(x; \chi_1) + x^{\beta_1}/\beta_1) - c(\chi)x^{\beta_1}/\beta_1$. The first part is covered by the bound just proved.

Case A: $f_2 = 1$. Then $c(\chi) = \mu/\varphi(Q)$, and the conductor q_1 of χ_1 divides Q , so $q_1 \leq Z$. The main terms combine to $\lambda \mu x / \varphi(Q)$ with $\lambda = 1 - x^{\beta_1-1}/\beta_1$. Since $1/\beta_1 \leq 1 + 2(1 - \beta_1)$, $1 - e^{-u} \geq (1 - e^{-1}) \min(u, 1)$ and $2(1 - \beta_1)e^{-u} = 2ue^{-u}/\log x \leq 2 \min(u, 1)/\log x$,

$$\lambda \geq 1 - e^{-u} - 2(1 - \beta_1)e^{-u} \geq (0.63 - 2/16) \min(u, 1) \geq \min(u, 1)/2.$$

So the error from Theorem 5.2 is at most $\lambda\mu x/(100\varphi(Q))$. By Theorem 5.1, $u \geq 16(1 - \beta_1) \geq 16/(c_2 Z^{1/2}(\log Z)^2)$, so $1/\lambda \leq 2 + c_2 Z^{1/2}(\log Z)^2/8 \leq C_5 Z$. Hence $|R_1| \leq 2AZ^3\mu/\varphi(Q) \leq \lambda\mu x/(100\varphi(Q))$ once $x \geq 200C_5AZ^4$. Altogether $S(x) \geq (1 - 2/100)\lambda\mu x/\varphi(Q) > 0$.

Case B: $f_2 > 1$. By (c), $|c(\chi)x^{\beta_1}/\beta_1| \leq (\mu/(4\varphi(Q))) \cdot 2x = \mu x/(2\varphi(Q))$, and $S(x) \geq (1 - 1/2 - 1/200 - 1/400)\mu x/\varphi(Q) > 0$.

In all cases $S(x) > 0$. So some prime $p \leq x$ with $p \nmid N_0$ and $p \equiv r \pmod{Q}$ has $B(p) > 0$, and then $W(p) > T$. $\square$

Remark 6.2 (Assessment). This is Linnik's theorem with weights. Every character coefficient is bounded by $\mathbb{E}_H|B|/\varphi(Q)$, not by the ℓ^1 mass of a cell representation. Theorem 5.2 controls all characters of conductor $\leq Q_G$ simultaneously, so the number of moduli d_i never multiplies the error. In [2] and in the first version of this note, each progression was instead treated separately with the Thorner–Zaman theorem [39]. That costs $\log x \gg K \max(\log Z, K)$ with $K = 1 + \log(\sum |c_i|/\varphi(d_i)/\mu)$, which here is of order $(\log T)^{14}$ instead of $(\log T)^7$ in the setting of the previous version. No lower bound for $1 - \beta_1$ is needed in Case B. In Case A we use the effective bound of Theorem 5.1 only to absorb the term R_1 .

7. AN ENERGY BOUND FOR AVOIDANCE INDICATORS

This section is self-contained and does not use arithmetic.

Setting. Let $\Omega = \prod_{v \in V_0} \Omega_v$ be a finite product of finite sets with a product probability measure $\pi = \bigotimes_v \pi_v$; write $\mathbb{P}$, $\mathbb{E}$ and $\|\cdot\| = \|\cdot\|_{L^2(\pi)}$. A *cylinder event* is a set $E = \{X_v = \sigma_v \mid (v \in \text{supp } E)\}$ with $\text{supp } E \subseteq V_0$ and $\sigma \in \prod_{\text{supp } E} \Omega_v$. For a finite family $\mathcal{E}$ of cylinder events (repetitions allowed) let $F = F_{\mathcal{E}} = \prod_{E \in \mathcal{E}} (1 - \mathbf{1}_E)$ be its *avoidance indicator*.

Efron–Stein decomposition (classical; [9, 21], [31, §8.3]). For $\varphi : \Omega \rightarrow \mathbb{R}$ and $W \subseteq V_0$ let $\mathbb{E}[\varphi \mid X_W]$ be the conditional expectation given the coordinates in W , and

$$\varphi^{=U} = \sum_{W \subseteq U} (-1)^{|U \setminus W|} \mathbb{E}[\varphi \mid X_W].$$

Then $\varphi = \sum_U \varphi^{=U}$, the components are mutually orthogonal, $\varphi^{=U}$ depends only on X_U , and $\mathbb{E}_u \varphi^{=U} = 0$ for $u \in U$, where $\mathbb{E}_u$ averages over the coordinate u . Put $L_u = I - \mathbb{E}_u$ and $L_V = \prod_{u \in V} L_u$; then $L_V \varphi = \sum_{U \subseteq V} \varphi^{=U}$. For a down-closed family $\mathcal{D}$ of subsets of V_0 , the function $\varphi^{\mathcal{D}} = \sum_{U \in \mathcal{D}} \varphi^{=U}$ is a linear combination of the $\mathbb{E}[\varphi \mid X_W]$, $W \in \mathcal{D}$ (by Möbius inversion), and

$$\mathbb{E}(\varphi - \varphi^{\mathcal{D}})^2 = \sum_{U \notin \mathcal{D}} \|\varphi^{=U}\|^2. \quad (7)$$

For $\mathcal{D} = \{U : |U| \leq t\}$ this is the *energy above level t* , $\text{En}(\varphi; t)$; it is the least $\mathbb{E}(\varphi - g)^2$ over sums g of functions of at most t coordinates.

Definition 7.1 (Energy generating function). For weights $\lambda_v \geq 1$ put $\mu_v = \lambda_v - 1$ and

$$G_\varphi(\lambda) = \sum_{U \subseteq V_0} \lambda^U \|\varphi^{=U}\|^2 = \sum_{V \subseteq V_0} \mu^V \|L_V \varphi\|^2.$$

The second form follows from the identity $\lambda^U = \sum_{V \subseteq U} \mu^V$ and $\|\varphi^{=U}\|^2 = \sum_{U \supseteq V} \|\varphi^{=V}\|^2$. For every τ ,

$$\sum_{U: \lambda^U > e^\tau} \|\varphi^{=U}\|^2 \leq e^{-\tau} G_\varphi(\lambda). \quad (8)$$

For a single event E with $\pi_E = \mathbb{P}(E)$ one computes $G_{1-\mathbf{1}_E} = 1 - 2\pi_E + \pi_E \prod_{v \in \text{supp } E} (\lambda_v - \mu_v \pi_v(\sigma_v))$, so for small $\pi_v(\sigma_v)$ the condition $\lambda^{\text{supp } E} \leq 2$ below is necessary for $G \leq 1$.

Hypergraphs. A hypergraph $\mathcal{H}$ is a finite set of subsets (*edges*) of V_0 . For $V \subseteq V_0$ put

$$N_{\mathcal{H}}(V) = \sum_{\mathcal{J} \subseteq \mathcal{H}: \bigcup_{J \in \mathcal{J}} (J \cap V) = V} (-1)^{|\mathcal{J}|}, \quad Q_\mu(\mathcal{H}) = \sum_{V \subseteq V_0} \mu^V N_{\mathcal{H}}(V)^2.$$

$N_{\mathcal{H}}(V)$ depends only on the traces $J \cap V$, and it is unchanged if edges are repeated or replaced by edges with the same trace: a class of $m \geq 1$ edges with equal traces contributes $\sum_{j \geq 1} \binom{m}{j} (-1)^j = -1$, exactly like one edge. If some edge misses V , then $N_{\mathcal{H}}(V) = 0$ (toggle that edge).

Lemma 7.2 (Cover bound; proved). *In the Setting, for every $V \subseteq V_0$,*

$$\|L_V F\|^2 \leq \mathbb{E}_x [N_{\mathcal{H}(x)}(V)^2], \quad \mathcal{H}(x) = \{\text{supp } E : E \in \mathcal{E}, x \in E\}.$$

Consequently $G_F(\lambda) \leq \mathbb{E}_x [Q_\mu(\mathcal{H}(x))]$ for all weights $\lambda_v \geq 1$.

Proof. Write $x = (\sigma, y)$ with $\sigma \in \Omega_V$, $y \in \Omega_{V_0 \setminus V}$. The operator L_V acts on the V -coordinates only, so $\|L_V F\|^2 = \mathbb{E}_y \|L_V g_y\|_{L^2(\Omega_V)}^2$, where $g_y = F(\cdot, y)$. Now g_y is the avoidance indicator on Ω_V of the restricted events: those $E \in \mathcal{E}$ whose conditions off V hold at y , restricted to $\text{supp } E \cap V$. Expand $g_y = \sum_{\mathcal{J}} (-1)^{|\mathcal{J}|} \mathbf{1}_{\mathcal{J}}$ over subfamilies $\mathcal{J}$ of restricted events, where $\mathbf{1}_{\mathcal{J}}$ is the indicator of their intersection, a cylinder on Ω_V (or empty). L_V kills every function that does not depend on all coordinates of V , hence every cylinder whose support is a proper subset of V . A cylinder with support V is a point indicator $\mathbf{1}_{\{\sigma\}}$. So $L_V g_y = L_V c_y$, where

$$c_y(\sigma) = \sum_{\mathcal{J}: \text{supp } \mathbf{1}_{\mathcal{J}} = V, \mathbf{1}_{\mathcal{J}} = \mathbf{1}_{\{\sigma\}}} (-1)^{|\mathcal{J}|}.$$

As L_V is an orthogonal projection in $L^2(\pi_V)$, $\|L_V c_y\|^2 \leq \mathbb{E}_\sigma c_y(\sigma)^2$. The subfamilies $\mathcal{J}$ counted in $c_y(\sigma)$ are exactly the subfamilies of events that hold at $x = (\sigma, y)$ whose traces on V cover V . Two events that hold at x and have the same support are the same cylinder, so by the remark above $c_y(\sigma) = N_{\mathcal{H}(x)}(V)$. Taking $\mathbb{E}_y$ gives the first claim. Multiply by μ^V and sum over V for the second. $\square$

The suppression is built in: $N_{\mathcal{H}(x)}(V) = 0$ as soon as some event that holds at x misses V .

For a finite multiset $\mathcal{C}$ of subsets of V_0 (possibly empty ones) put

$$\Theta_\lambda(\mathcal{C}) = \sum_{\mathcal{J} \subseteq \mathcal{C}} (-1)^{|\mathcal{J}|} \lambda^{\bigcup \mathcal{J}} \quad (\Theta_\lambda(\emptyset) = 1).$$

Lemma 7.3 (Polarization; proved). *Let $\lambda_v \geq 1$, $\mu_v = \lambda_v - 1$, and let $P \subseteq V_0$ be random with $\mathbb{P}(v \in P) = \mu_v/\lambda_v$ independently. For every hypergraph $\mathcal{H}$,*

$$Q_\mu(\mathcal{H}) = \mathbb{E}_P[\Theta_\lambda(\mathcal{H}_P)^2], \quad \mathcal{H}_P = \{J \in \mathcal{H} : J \cap P = \emptyset\}.$$

Proof. Let $\tau(R) = \mathbf{1}[R \cap J \neq \emptyset \ \forall J \in \mathcal{H}]$ for $R \subseteq V_0$, and $\hat{\tau}(V) = \sum_{R \subseteq V} (-1)^{|V \setminus R|} \tau(R)$. Expanding $\tau(R) = \sum_{\mathcal{J} \subseteq \mathcal{H}} (-1)^{|\mathcal{J}|} \mathbf{1}[R \cap \bigcup \mathcal{J} = \emptyset]$, and using that the Möbius transform of $R \mapsto \mathbf{1}[R \cap U = \emptyset]$ is $V \mapsto (-1)^{|V|} \mathbf{1}[V \subseteq U]$, we get $\hat{\tau}(V) = (-1)^{|V|} N_{\mathcal{H}}(V)$. Let

$$\mathcal{T}(p) = \sum_V \hat{\tau}(V) p^V = \sum_R \tau(R) \prod_{v \in R} p_v \prod_{v \notin R} (1 - p_v)$$

be the multilinear extension of τ . If the p_v are independent with mean 0 and variance μ_v , distinct monomials are orthogonal, and $\mathbb{E}[\mathcal{T}(p)^2] = \sum_V \mu^V \hat{\tau}(V)^2 = Q_\mu(\mathcal{H})$. Take $p_v = 1$ with probability μ_v/λ_v and $p_v = -\mu_v$ otherwise (mean $\mu_v/\lambda_v - \mu_v/\lambda_v = 0$, variance $\mu_v/\lambda_v + \mu_v^2/\lambda_v = \mu_v$), and $P = \{v : p_v = 1\}$. Only the $R \supseteq P$ contribute; writing $R = P \cup B$ with $B \subseteq P^c$,

$$\mathcal{T}(p) = \sum_{B \subseteq P^c} (-\mu)^B \lambda^{P^c \setminus B} \tau(P \cup B).$$

Now $\tau(P \cup B) = \sum_{\mathcal{J} \subseteq \mathcal{H}_P} (-1)^{|\mathcal{J}|} \mathbf{1}[B \cap \bigcup \mathcal{J} = \emptyset]$, and for $U \subseteq P^c$,

$$\sum_{B \subseteq P^c \setminus U} (-\mu)^B \lambda^{P^c \setminus B} = \lambda^U \prod_{v \in P^c \setminus U} (\lambda_v - \mu_v) = \lambda^U.$$

Hence $\mathcal{T}(p) = \Theta_\lambda(\mathcal{H}_P)$. $\square$

Lemma 7.4 (Matching bound; proved). *If every $J \in \mathcal{C}$ has $w_J := \lambda^J \leq 2$, then for every matching $\mathcal{M} \subseteq \mathcal{C}$ (pairwise disjoint members)*

$$|\Theta_\lambda(\mathcal{C})| \leq \prod_{J \in \mathcal{M}} (w_J - 1); \quad \text{in particular } |\Theta_\lambda(\mathcal{C})| \leq 1.$$

Proof. For $v \in V_0$ let $\mathcal{C} - v$ be the members not containing v and $\mathcal{C}/v = \{J \setminus \{v\} : J \in \mathcal{C}\}$. Splitting $\mathcal{J}$ according to whether it contains a member through v ,

$$\Theta(\mathcal{C}) = \Theta(\mathcal{C} - v) + \lambda_v [\Theta(\mathcal{C}/v) - \Theta(\mathcal{C} - v)] = \lambda_v \Theta(\mathcal{C}/v) - \mu_v \Theta(\mathcal{C} - v).$$

Both families again have all weights at most 2. We induct on $|\bigcup \mathcal{C}|$. If $\emptyset \in \mathcal{C}$, then $\Theta(\mathcal{C}) = 0$ (toggling the empty member pairs the terms), and the claim holds. If $\mathcal{C} = \emptyset$, then $\Theta = 1$ and $\mathcal{M} = \emptyset$. This covers the base case $\bigcup \mathcal{C} = \emptyset$. If $\mathcal{M} = \emptyset$ and $\mathcal{C} \neq \emptyset$, it suffices to prove the bound for $\mathcal{M} = \{J\}$ with any $J \in \mathcal{C}$, as $w_J - 1 \leq 1$. So let $J_0 \in \mathcal{M}$ be nonempty, $v \in J_0$, and $\Pi' =$

$\prod_{J \in \mathcal{M} \setminus \{J_0\}} (w_J - 1)$. In $\mathcal{C}/v$ the family $(\mathcal{M} \setminus \{J_0\}) \cup \{J_0 \setminus \{v\}\}$ is a matching, so $|\Theta(\mathcal{C}/v)| \leq (w_{J_0}/\lambda_v - 1)\Pi'$. In $\mathcal{C} - v$ the family $\mathcal{M} \setminus \{J_0\}$ is a matching, so $|\Theta(\mathcal{C} - v)| \leq \Pi'$. Therefore $|\Theta(\mathcal{C})| \leq (w_{J_0} - \lambda_v)\Pi' + \mu_v\Pi' = (w_{J_0} - 1)\Pi'$. $\square$

Theorem 7.5 (Energy bound; proved). (a) *If every edge of a hypergraph $\mathcal{H}$ has $\lambda^J \leq 2$, then $Q_\mu(\mathcal{H}) \leq \prod_{J \in \mathcal{M}} (\lambda^J - 1)$ for every matching $\mathcal{M} \subseteq \mathcal{H}$; in particular $Q_\mu(\mathcal{H}) \leq 1$.* (b) *In the Setting, if $\lambda_v \geq 1$ and $\lambda^{\text{supp } E} \leq 2$ for every $E \in \mathcal{E}$, then*

$$G_F(\lambda) = \sum_U \lambda^U \|F^{=U}\|^2 \leq 1.$$

Proof. (a) By Lemma 7.3 and Lemma 7.4 applied to $\mathcal{H}_P$ with the matching $\mathcal{M} \cap \mathcal{H}_P$, $Q_\mu(\mathcal{H}) \leq \mathbb{E}_P \prod_{J \in \mathcal{M} \cap \mathcal{H}_P} (\lambda^J - 1)^2$. The members of $\mathcal{M}$ are disjoint, so the events $\{J \cap P = \emptyset\}$ are independent, with probability $\prod_{v \in J} \lambda_v^{-1} = 1/w_J$. Hence $Q_\mu(\mathcal{H}) \leq \prod_{J \in \mathcal{M}} [(1 - 1/w_J) + (w_J - 1)^2/w_J] = \prod_{J \in \mathcal{M}} (w_J - 1)$. (b) Lemma 7.2 and (a), since $\mathcal{H}(x)$ consists of supports of events. $\square$

Corollary 7.6 (Energy tail; proved). *In the Setting, if every event of $\mathcal{E}$ has $|\text{supp } E| \leq k$ ($k \geq 1$), then for $f \in \{F, 1 - F\}$ and every integer $t \geq 0$,*

$$\text{En}(f; t) = \sum_{|U| > t} \|f^{=U}\|^2 \leq 2^{-(t+1)/k}.$$

Proof. Theorem 7.5(b) with $\lambda_v = 2^{1/k}$ and (8); $(1 - F)^{=U} = -F^{=U}$ for $U \neq \emptyset$. $\square$

Remark 7.7 (Sharpness; proved). Over general product spaces the base $2^{-1/k}$ cannot be improved. Take m events on pairwise disjoint sets of k coordinates, each fixing at every coordinate of its support a value of probability p ; put $\pi = p^k$. As the factors $1 - \mathbf{1}_{E_i}$ are independent, $F^{=U} = \prod_i (1 - \mathbf{1}_{E_i})^{=U \cap \text{supp } E_i}$. For U the union of the supports of j of the events this gives $\|F^{=U}\|^2 = [\pi(1 - p)^k]^j (1 - \pi)^{2(m-j)}$, and $|U| = jk$. Hence

$$\text{En}(F; jk - 1) \geq \binom{m}{j} [\pi(1 - p)^k]^j (1 - \pi)^{2(m-j)} \rightarrow e^{-2s} \frac{s^j}{j!}$$

as $p \rightarrow 0$ with $m = \lfloor s/\pi \rfloor$. With $s = j/2$ and $j! \leq e j^{j+1/2} e^{-j}$, the limit is at least $2^{-j}/(e\sqrt{j}) = 2^{-(t+1)/k}/(e\sqrt{j})$ for $t = jk - 1$. So no bound $C\rho^{-(t+1)/k}$ with $\rho > 2$ holds for all such systems (with alphabets of size about $1/p$), and at most a factor of order $\sqrt{t/k}$ could be gained. The condition $\lambda^{\text{supp } E} \leq 2$ in Theorem 7.5 is necessary, by the single-event formula above.

Remark 7.8 (Boolean form and comparison; proved, except the cited constant). A width- k DNF $g : \{\pm 1\}^n \rightarrow \{\pm 1\}$ is $1 - 2F$ for the family of its terms (a term with a literal set is a cylinder event). As $g^{=U} = -2F^{=U}$ for $U \neq \emptyset$, Corollary 7.6 gives, for every product measure on $\{\pm 1\}^n$ (Efron–Stein weights; Fourier weights for the uniform measure),

$$\sum_{|S| > t} \hat{g}(S)^2 \leq 4 \cdot 2^{-(t+1)/k},$$

i.e. g is ε -concentrated up to degree $k \log_2(4/\varepsilon)$. The switching-lemma route of Linial, Mansour and Nisan gives $\sum_{|S| \geq t} \hat{g}(S)^2 \leq 2 \cdot 2^{-t/(20k)}$ [31, §4.4], [24]. Also $\lambda^{|U|} \geq 1 + |U| \ln \lambda$ gives total influence $\sum_U |U| \|F^{=U}\|^2 \leq (k/\ln 2) \mathbb{P}(F = 0)$. The same bounds hold for q -ary DNFs with literals $x_v \in S_v$ (split each term into single-value cylinders on the same support; F and the supports do not change). We have not found Theorem 7.5 in the literature; see the introduction for what we checked.

Digit filtrations. For the application we need weights that grow along a filtration of each coordinate. Suppose that each coordinate X_ℓ ($\ell \in \mathcal{P}$, a finite index set) is a tuple of independent *digits* $(\delta_{\ell,i})_{i_0(\ell) \leq i < f_\ell}$, with arbitrary distributions, and that the coordinates are independent; the Efron–Stein decomposition $F = \sum_U F^{=U}$ below is taken over the digits, so U is a set of pairs (ℓ, i) . For such U write $U_\ell = \{i : (\ell, i) \in U\}$. We say that an event E is a *digit-prefix event* if, for some integers $v_\ell(E) \geq 0$, it fixes exactly the digits $\delta_{\ell,i}$ with $i_0(\ell) \leq i < v_\ell(E)$, for each ℓ (to arbitrary values). Such an event is a cylinder event on the digits.

Theorem 7.9 (Filtration energy bound; proved). *In this situation let $\lambda_\ell \geq 1$ ($\ell \in \mathcal{P}$), and suppose that every event of $\mathcal{E}$ is a digit-prefix event with $\prod_\ell \lambda_\ell^{2v_\ell(E)} \leq 2$. Then*

$$\sum_U \left(\prod_{\ell: U_\ell \neq \emptyset} \lambda_\ell^{1+\max U_\ell} \right) \|F^{=U}\|^2 \leq 1.$$

Proof. Write $\Lambda_{\ell,i} = \lambda_\ell^{i+1}$, $\Lambda_{\ell,i_0(\ell)-1} = 1$, and $\mu'_{\ell,j} = \Lambda_{\ell,j} - \Lambda_{\ell,j-1} \geq 0$ for $i_0(\ell) \leq j < f_\ell$. For such j let $P_{\ell,j} = I - \mathbb{E}_{B_{\ell,j}}$, where $\mathbb{E}_{B_{\ell,j}}$ averages over the *tail block* $B_{\ell,j} = (\delta_{\ell,i})_{i \geq j}$. Averaging over a digit in U kills $F^{=U}$, and $F^{=U}$ does not depend on digits outside U ; hence $P_{\ell,j} F^{=U} = F^{=U}$ if $U_\ell \neq \emptyset$ and $\max U_\ell \geq j$, and $P_{\ell,j} F^{=U} = 0$ otherwise. Call a set V of pairs (ℓ, j) a *selection* if it contains at most one pair for each ℓ , and put $L'_V = \prod_{(\ell,j) \in V} P_{\ell,j}$. Then $\|L'_V F\|^2 = \sum_{U \succ V} \|F^{=U}\|^2$, where $U \succ V$ means $\max U_\ell \geq j$ for all $(\ell, j) \in V$. For fixed U , the sum of μ'^V over the selections $V \prec U$ factorises as $\prod_\ell (1 + \sum_{i_0(\ell) \leq j \leq \max U_\ell} \mu'_{\ell,j}) = \prod_{\ell: U_\ell \neq \emptyset} \Lambda_{\ell, \max U_\ell}$. So the left side of the claim equals $\sum_V \mu'^V \|L'_V F\|^2$, the sum over selections.

Fix a selection V and pass to the coarser product space whose coordinates are, for $(\ell, j) \in V$, the head $(\delta_{\ell,i})_{i < j}$ and the tail block $B_{\ell,j}$, and, for the other ℓ , the whole X_ℓ . There L'_V is the operator $L_{V'}$ of the Setting for the set V' of tail blocks. Each event E is a disjoint union of cylinder events on the coarse coordinates with the same coarse support (enumerate the unfixed digits of each coarse coordinate that E involves), and splitting every event in this way does not change F . Exactly one piece of E holds at x if E does. The trace of the coarse support of E on V' is the set of $B_{\ell,j}$ with $(\ell, j) \in V$ and $j < v_\ell(E)$. Identify $B_{\ell,j}$ with the pair (ℓ, j) and put $\hat{E} = \{(\ell, j) : i_0(\ell) \leq j < v_\ell(E)\}$.

By Lemma 7.2 and the remark on traces,

$$\|L'_V F\|^2 \leq \mathbb{E}_x[N_{\hat{\mathcal{H}}(x)}(V)^2], \quad \hat{\mathcal{H}}(x) = \{\hat{E} : E \in \mathcal{E}, x \in E\}.$$

Summing over selections and adding the nonnegative terms of all other sets V of pairs, the left side of the claim is at most $\mathbb{E}_x[Q_{\mu'}(\hat{\mathcal{H}}(x))]$. By Theorem 7.5(a) it remains to show $\prod_{(\ell,j) \in \hat{E}} (1 + \mu'_{\ell,j}) \leq 2$ for every event. Fix ℓ and write $\lambda = \lambda_\ell$, $v = v_\ell(E)$, $i_0 = i_0(\ell)$, and assume $v > i_0$. Here $1 + \mu'_{\ell,i_0} = \lambda^{i_0+1}$ and $1 + \mu'_{\ell,j} = 1 + \lambda^j(\lambda - 1)$ for $j > i_0$. The hypothesis gives $\lambda^{2v} \leq 2$, so $\lambda^j \leq \lambda^v \leq \sqrt{2} < \lambda + 1$ for $j < v$, i.e. $1 + \lambda^j(\lambda - 1) \leq \lambda^2$. The factor of ℓ is therefore at most $\lambda^{i_0+1} \lambda^{2(v-i_0)} \leq \lambda^{2v}$, and the product over ℓ is at most $\prod_\ell \lambda_\ell^{2v_\ell(E)} \leq 2$. $\square$

Corollary 7.10 (Modulus-weighted tail; proved). *In Theorem 7.9 let the $\ell \in \mathcal{P}$ be primes and $\lambda_\ell = 2^{\log \ell / (2\mathcal{L})}$, and suppose $\prod_\ell \ell^{v_\ell(E)} \leq T$ for every event. For a set U of digits put $m_U = \prod_{\ell: U_\ell \neq \emptyset} \ell^{1+\max U_\ell}$. Then for every $\tau \geq 0$*

$$\sum_{U: m_U > e^\tau} \|F^{=U}\|^2 \leq 2^{-\tau/(2\mathcal{L})}.$$

Proof. $\prod_\ell \lambda_\ell^{2v_\ell(E)} = 2^{\log \prod_\ell \ell^{v_\ell(E)} / \mathcal{L}} \leq 2$, and the weight of U in Theorem 7.9 is $2^{\log m_U / (2\mathcal{L})}$. $\square$

The cost of a junta is thus measured by the modulus m_U a function of the digits in U needs, not by the number of coordinates. For the Erdős–Straus events (digits in base ℓ) this removes both the binary width and the support size from the junta modulus.

8. THE RESIDUAL SYSTEM AND THE SANDWICH MINORANT

Throughout this section T is large, $\beta = 1 + 1/\log \mathcal{L}$, $\eta = \frac{3}{4} \log \beta$, and (Q, r) is the square-class quarantine of Theorem 3.3; $E_1, \dots, E_m$ ($m \leq T^2$) are the events of the surviving atoms, in some order, $A_i = \mathbf{1}_{E_i}$, $F_{<i} = \prod_{j<i} (1 - A_j)$, $F = F_{<m+1}$, $S = \sum_i \mathbb{P}(E_i) \leq S_\beta$ and $\delta = \mathbb{E}F$. By (4) and Lemma 2.8, every subfamily of $\{E_i\}$ satisfies the hypothesis of Lemma 2.8, $\delta \geq e^{-4S_\beta/3}$, and for T large

$$\eta \leq 0.19 \quad (\text{as soon as } \mathcal{L} \geq 33). \quad (9)$$

Digits. For $\ell \in \mathcal{P}$ write $x \in \Omega_\ell$ in base ℓ as $x = \sum_{i < f_\ell} \delta_{\ell,i} \ell^i$. The digits $\delta_{\ell,i}$ with $i < a_\ell$ are those of r , hence constant on Ω_ℓ . The others are independent and uniform: $\delta_{\ell,0}$ on $\{1, \dots, \ell - 1\}$ if $a_\ell = 0$, and $\delta_{\ell,i}$ on $\{0, \dots, \ell - 1\}$ for $\max(a_\ell, 1) \leq i < f_\ell$; indeed $x \mapsto (\delta_{\ell,i})_{i \geq a_\ell}$ is a bijection from Ω_ℓ onto the product of these ranges. So the free digits are those with $i \geq i_0(\ell) := a_\ell$, and $E_{M,D}$ is a digit-prefix event with $v_\ell(E_{M,D}) = v_\ell(M)$ for $\ell \in \text{supp}(M, D)$ and $v_\ell(E_{M,D}) = 0$ otherwise. A function of the digits $\delta_{\ell,i}$ with $i \leq i_\ell$ depends only on X_ℓ modulo ℓ^{1+i_ℓ} .

Cells. A *consistent unit cell* is a class $b \bmod d$ with d composed of primes of $\mathcal{P}$, $d \mid \prod_{\ell \in \mathcal{P}} \ell^{f_\ell}$, $\gcd(b, d) = 1$ and $b \equiv r \pmod{\gcd(d, Q)}$; on Ω it is the

event $\{X_\ell \equiv b \pmod{\ell^{v_\ell(d)}} \forall \ell \mid d\}$, and on $\mathbb{Z}$ the class itself. The events $E_{M,D}$ are consistent unit cells (by survival). The intersection of two consistent unit cells is empty or a consistent unit cell whose modulus is the lcm of the two, both on Ω and on $\mathbb{Z}$. A function on Ω that depends only on $(X_\ell \bmod \ell^{b_\ell})_\ell$ is a linear combination of consistent unit cells of modulus $\prod_\ell \ell^{b_\ell}$ (the indicators of the points of its level sets).

Lemma 8.1 (Haar means; proved). *Let $Q' = Q\ell'$ with ℓ' a prime not dividing $Q \prod_{\ell \in \mathcal{P}} \ell$, and r' the class modulo Q' with $r' \equiv r \pmod{Q}$ and $r' \equiv 1 \pmod{\ell'}$ (or $Q' = Q$, $r' = r$). Let φ be a combination of consistent unit cells with lcm of moduli D . With H as in Theorem 6.1 for Q' , r' and D , $\mathbb{E}_H \varphi = \mathbb{E} \varphi$. The same holds for $\varphi\psi$, where ψ is a character modulo a squarefree $f \mid D$ coprime to Q , viewed on Ω through $X_\ell \bmod \ell$ for $\ell \mid f$.*

Proof. Let $N_0 = \text{lcm}(Q', D)$. By the Chinese remainder theorem the uniform measure on H is the product of the uniform measures on $H_p = \{x \in (\mathbb{Z}/p^{v_p(N_0)})^\times : x \equiv r' \pmod{p^{v_p(Q')}}\}$. For $\ell \in \mathcal{P}$ with $\ell \mid D$, $v_\ell(Q') = a_\ell$ and $v_\ell(N_0) \leq f_\ell$, and reduction maps the uniform measure on Ω_ℓ to the uniform measure on H_ℓ (all fibres have the same size). The other primes of N_0 do not divide D , and φ, ψ do not depend on them. $\square$

For $j \leq m$ let $F^{(j)}$ be the function of the digits not fixed by E_j obtained from $F_{<j}$ by giving the digits fixed by E_j their values on E_j . Then $A_j F_{<j} = A_j F^{(j)}$. At ℓ , the digits not fixed by E_j are those with $i \geq i_0^{(j)}(\ell) := \max(a_\ell, v_\ell(E_j))$, and $F^{(j)}$ is the avoidance indicator of the restrictions of the events E_i ($i < j$) compatible with E_j . Each restriction fixes the digits $i_0^{(j)}(\ell) \leq i < v_\ell(E_i)$, so it is a digit-prefix event for the filtration starting at $i_0^{(j)}$, with the same $v_\ell(E_i)$ and $\prod_\ell \ell^{v_\ell(E_i)} \leq M_i \leq T$. (A restriction may fix nothing; then $F^{(j)} \equiv 0$.)

Lemma 8.2 (Bazzi–Razborov sandwich [3, 35]; proved). *Let $u_1, \dots, u_m$ be arbitrary real functions, $v_i = \sum_{j < i} A_j u_j$, and*

$$B = 1 - \sum_{i \leq m} A_i (1 - v_i)^2.$$

Then $B \leq F$ pointwise, and with $e_j = F_{<j} - u_j$,

$$F - B = \sum_i A_i \left(\sum_{j < i} A_j e_j \right)^2, \quad \mathbb{E}[F - B] \leq m^2 \sum_{j \leq m} \mathbb{E}[A_j e_j^2].$$

If u_j is a function of the digits not fixed by E_j , then $\mathbb{E}[A_j e_j^2] = \mathbb{P}(E_j) \mathbb{E}(F^{(j)} - u_j)^2$.

Proof. The identity $1 - F = \sum_i A_i F_{<i}$ holds pointwise: the sum counts the first occurring event. Put $f_{<i} = 1 - F_{<i} = \sum_{j < i} A_j F_{<j}$. If $F_{<i} = 1$, then $A_j = 0$ for all $j < i$, so $v_i = 0$ and $A_i (1 - v_i)^2 = A_i = A_i F_{<i}$. If

$F_{<i} = 0$, then $f_{<i} = 1$, and $A_i(1 - v_i)^2 - A_i F_{<i} = A_i(f_{<i} - v_i)^2$. In both cases $A_i(1 - v_i)^2 - A_i F_{<i} = A_i(f_{<i} - v_i)^2$, and summing over i ,

$$F - B = \sum_i A_i[(1 - v_i)^2 - F_{<i}] = \sum_i A_i(f_{<i} - v_i)^2 \geq 0,$$

with $f_{<i} - v_i = \sum_{j < i} A_j(F_{<j} - u_j)$. By Cauchy–Schwarz, $(\sum_{j < i} A_j e_j)^2 \leq m \sum_{j < i} A_j e_j^2$. Drop $A_i \leq 1$ and sum over i . For the last claim, $A_j e_j^2 = A_j(F^{(j)} - u_j)^2$, and $F^{(j)} - u_j$ is independent of A_j . $\square$

This is Razborov’s simplification [35] of Bazzi’s argument [3]. The choice of u_j as an ℓ^2 -approximation of the “conditional survival function” $F^{(j)}$ is the one used there. We include the short proof for completeness. The identity and the inequality $B \leq F$ use only that each A_i takes values in $\{0, 1\}$, so they also hold for the functions on $\mathbb{Z}$ obtained from cell expansions.

Choice of u_j . Fix $\tau \geq 0$. For $j \leq m$ let $\mathcal{D}_j$ be the family of sets U of digits not fixed by E_j with $m_U \leq e^\tau$, where $m_U = \prod_{\ell: U_\ell \neq \emptyset} \ell^{1+\max U_\ell}$; it is down-closed. Put

$$u_j = (F^{(j)})^{\mathcal{D}_j} = \sum_{U \in \mathcal{D}_j} (F^{(j)})^{=U}, \quad (10)$$

the Efron–Stein decomposition being taken over the digits not fixed by E_j .

Lemma 8.3 (Approximation error; proved). *With u_j as in (10), $\mathbb{E}[F - B] \leq m^2 S 2^{-\tau/(2\mathcal{L})}$.*

Proof. By (7) and Corollary 7.10, applied to $F^{(j)}$ with the filtration starting at $i_0^{(j)}$ (the digits are indexed by their absolute position, so m_U is the modulus needed), $\mathbb{E}(F^{(j)} - u_j)^2 \leq 2^{-\tau/(2\mathcal{L})}$. Conclude with Lemma 8.2. $\square$

Lemma 8.4 (Cell structure of B ; proved). *With u_j as in (10), B is a finite real combination of consistent unit cells of modulus at most $T^3 e^{2\tau}$, composed of primes of $\mathcal{P}$. The identity $B = 1 - \sum_i A_i(1 - v_i)^2$ also holds for the expanded cell combination evaluated at any integer n .*

Proof. By (7), u_j is a combination of the functions $\mathbb{E}[F^{(j)} \mid \delta_W]$, $W \in \mathcal{D}_j$. Such a function depends only on $X_\ell \bmod \ell^{1+\max W_\ell}$ for the ℓ with $W_\ell \neq \emptyset$, so it is a combination of consistent unit cells of modulus $m_W \leq e^\tau$. Expanding,

$$B = 1 - \sum_i A_i + 2 \sum_i \sum_{j < i} A_i A_j u_j - \sum_i \sum_{j, j' < i} A_i A_j A_{j'} u_j u_{j'}.$$

Each term is a product of at most three events (moduli $\leq T$) and at most two u ’s, so every cell is an intersection of consistent unit cells of total modulus at most $T^3 e^{2\tau}$. The last claim holds because the intersection identities hold on $\mathbb{Z}$. $\square$

Lemma 8.5 (ℓ^1 -tightness; proved). *If $B \leq F$ pointwise, $F \geq 0$ and $\mathbb{E}[F - B] \leq \eta \mathbb{E}B$, then $\mathbb{E}|B| \leq (1 + 2\eta)\mathbb{E}B$.*

Proof. $B^- := \max(-B, 0) \leq F - B$ since $F \geq 0$, and $\mathbb{E}|B| = \mathbb{E}B + 2\mathbb{E}B^-$. $\square$

Lemma 8.6 (Twist; proved). *Assume $\mathbb{E}[F - B] \leq \delta/100$, and write B as in Lemma 8.4, with $\mu = \mathbb{E}B$. Let ψ be a real primitive character of conductor $f > 1$ with $\gcd(f, Q) = 1$ and $f \mid d_i$ for some cell modulus d_i . Then $\mu \geq 0.99\delta$ and $|\mathbb{E}[B\psi]| \leq \mu/4$, where ψ is viewed on Ω through $X_\ell \bmod \ell$, $\ell \mid f$.*

Proof. $\mu = \mathbb{E}F - \mathbb{E}[F - B] \geq 0.99\delta$. As $2 \nmid f$, f is squarefree. Its prime factors divide d_i , so they lie in $\mathcal{P}$, and $a_\ell = 0$ for them as $\gcd(f, Q) = 1$. So $\psi = \prod_{\ell \mid f} \psi_\ell(X_\ell \bmod \ell)$ with nontrivial characters ψ_ℓ of the uniformly distributed digits $\delta_{\ell,0}$. Now

$$|\mathbb{E}[B\psi]| \leq \mathbb{E}|B - F| + |\mathbb{E}[F\psi]| = \mathbb{E}[F - B] + |\mathbb{E}[F\psi]|.$$

Fix $\ell_0 \mid f$ and let F' be the avoidance indicator of the events E_i with $\ell_0 \notin \text{supp } E_i$. Given the other coordinates $X_{-\ell_0}$, we have $F = F' \cdot \mathbf{1}[X_{\ell_0} \notin \text{Forb}]$, where Forb is the union of the classes $X_{\ell_0} \equiv -4D_i \pmod{\ell_0^{v_{\ell_0}(M_i)}}$ over the events $E_i \ni \ell_0$ whose other conditions hold. Write $E_i \setminus \ell_0$ for the event given by those other conditions, and π_i for the Ω_{ℓ_0} -probability of the class at ℓ_0 , so that $\mathbb{P}(E_i) = \mathbb{P}(E_i \setminus \ell_0) \pi_i$. Since $\mathbb{E}_{X_{\ell_0}} \psi_{\ell_0} = 0$,

$$\begin{aligned} |\mathbb{E}[F\psi]| &= \left| \mathbb{E} \left[F' \prod_{\ell \neq \ell_0} \psi_\ell \mathbb{E}_{X_{\ell_0}} [\psi_{\ell_0} \mathbf{1}[X_{\ell_0} \in \text{Forb}]] \right] \right| \\ &\leq \sum_{i: \ell_0 \in \text{supp } E_i} \mathbb{P}(F' = 1, E_i \setminus \ell_0) \pi_i. \end{aligned}$$

The event $E_i \setminus \ell_0$ is determined by the $s(E_i) - 1$ coordinates of $\text{supp } E_i \setminus \{\ell_0\}$, so Lemma 2.8, applied to the family defining F' , gives $\mathbb{P}(E_i \setminus \ell_0 \mid F' = 1) \leq \beta^{s(E_i)-1} \mathbb{P}(E_i \setminus \ell_0)$. Hence

$$|\mathbb{E}[F\psi]| \leq \beta^{-1} \sum_{i: \ell_0 \in \text{supp } E_i} \beta^{s(E_i)} \mathbb{P}(E_i) \mathbb{E}F' = \beta^{-1} \tilde{w}_{\ell_0} \mathbb{E}F' \leq \eta \mathbb{E}F'.$$

The same bound gives $\mathbb{E}F' - \mathbb{E}F = \mathbb{E}[F' \mathbf{1}[X_{\ell_0} \in \text{Forb}]] \leq \eta \mathbb{E}F'$, so $\mathbb{E}F' \leq \delta/(1 - \eta)$. By (9), $|\mathbb{E}[B\psi]| \leq (0.01 + \eta/(1 - \eta))\delta \leq 0.245\delta \leq 0.245\mu/0.99 < \mu/4$. $\square$

9. PROOF OF THE MAIN THEOREMS

Theorem 9.1 (Assembly; proved modulo Theorems 5.2 and 3.1). *Let T be large, let (Q, r) be the square-class quarantine of Theorem 3.3, put $S_1 = C\mathcal{L}^3 \log \mathcal{L}$ (the bound for S_β in (4)) and*

$$\tau = 2\mathcal{L} \lceil \log_2 (100 T^4 (S_1 + 1) e^{3S_1}) \rceil.$$

Then there is a hard prime p with $p > T$, $W(p) > T$ and $\log p \leq C_6 \log Z$, where C_6 is absolute and

$$\log Z \leq \log Q + 4\tau + 6\mathcal{L} + 1.$$

Proof. Take the residual system of (Q, r) ; $m \leq T^2$. If $m = 0$, then $F \equiv 1$; we put $B = 1$ (one cell of modulus 1), so that $\mu = A = 1$ and the twist condition is vacuous, and go directly to the transfer step below. Otherwise

define B by Lemma 8.2 with u_j as in (10). By Lemma 8.3, the choice of τ , $S \leq S_\beta \leq S_1$ and $\delta \geq e^{-4S_\beta/3} \geq e^{-3S_1}$,

$$\mathbb{E}[F - B] \leq m^2 S 2^{-\tau/(2\mathcal{L})} \leq \frac{S e^{-3S_1}}{100(S_1 + 1)} \leq \frac{e^{-3S_1}}{100} \leq \frac{\delta}{100}.$$

Lemma 8.6 gives $\mu = \mathbb{E}B \geq 0.99\delta > 0$, and, as $\mathbb{E}[F - B] \leq \mu/99$, Lemma 8.5 gives $\mathbb{E}|B| \leq 1.03\mu$. By Lemma 8.4, B is a combination of consistent unit cells of moduli $d_i \leq T^3 e^{2\tau}$.

Pointwise inequality on integers. For every integer n the values $A_i(n) \in \{0, 1\}$ satisfy the hypotheses of Lemma 8.2, and the cell expansion of B is valid on $\mathbb{Z}$ (Lemma 8.4). So $B(n) \leq F(n)$, and by Lemma 2.5(i), $B(n) \leq \mathbf{1}[W(n) > T]$ for every $n \equiv r \pmod{Q}$.

Transfer. Let $R = \max(T, \max_i d_i)$, choose a prime $\ell_{\text{aux}} \in (R, 2R]$, put $Q' = Q\ell_{\text{aux}}$, and let r' be the class modulo Q' with $r' \equiv r \pmod{Q}$ and $r' \equiv 1 \pmod{\ell_{\text{aux}}}$. Then $8 \mid Q'$, $r' \equiv 1 \pmod{8}$, and r' is a quadratic residue modulo every odd prime of Q' . As $\ell_{\text{aux}} > \max d_i$ is prime, $\gcd(d_i, Q') = \gcd(d_i, Q)$, so the cells are consistent for (Q', r') , and no cell or event involves ℓ_{aux} . The inequality $B \leq \mathbf{1}[W > T]$ holds on $n \equiv r' \pmod{Q'}$. The functions B and $|B|$ depend only on $X \bmod D$, so both are combinations of consistent unit cells (point cells modulo D), and by Lemma 8.1, $\mathbb{E}_H B = \mu$ and $\mathbb{E}_H |B|/\mu \leq 1.03 \leq Z^{1/4}$. A real primitive character of conductor $f > 1$ coprime to Q' is coprime to Q , so Lemmas 8.6 and 8.1 give the twist condition. Here $\log Z \leq \log Q + \log(2R) + \log \max d_i \leq \log Q + \log 2 + 2(3\mathcal{L} + 2\tau)$. Theorem 6.1 with $\log x = C_2(1 + \log 1.03) \log Z$ gives a prime $p \equiv r' \pmod{Q'}$ with $W(p) > T$ and $\log p \leq C_6 \log Z$. Since $\ell_{\text{aux}} \mid p - 1$, $p > \ell_{\text{aux}} > T$; and $p \equiv r \pmod{Q}$ with $840 \mid Q$, so p is hard by Lemma 2.5(iii). $\square$

The prime ℓ_{aux} is needed for infinitely many *distinct* primes: $W(p) > T$ alone does not force $p > T$.

Proof of Theorem 1.1. By (4), $\tau \ll \mathcal{L}(\mathcal{L} + S_1) \ll \mathcal{L}^4 \log \mathcal{L}$ and $\log Q \ll \mathcal{L}^3(\log \mathcal{L})^5$. So $\log Z \ll \mathcal{L}^4 \log \mathcal{L}$, and Theorem 9.1 gives a hard prime $p > T$ with $W(p) > T$ and $\log p \leq C\mathcal{L}^4 \log \mathcal{L}$. As $\mathcal{L} < \log p$, this gives $\mathcal{L}^4 \geq \log p / (C \log \log p)$, i.e. $W(p) > T \geq \exp(c(\log p)^{1/4}(\log \log p)^{-1/4})$. Applying this to a sequence $T_1 < T_2 < \dots$ with T_{j+1} larger than the prime found for T_j gives infinitely many distinct primes. $\square$

Remark 9.2 (Where the exponent comes from; Assessment). The certified bound is $\log p \ll \log Z \ll \log Q + \tau$. The quarantine term is $\log Q \ll \mathcal{L}^3(\log \mathcal{L})^5$, of the order of the Haar density (Theorems 3.3 and 4.4). The bottleneck is the junta term $\tau \asymp \mathcal{L}(S_1 + \mathcal{L}) \asymp \mathcal{L}^4 \log \mathcal{L}$: the cells of the sandwich have modulus about $e^{\mathcal{L}S}$. In the previous version both terms were of order $\mathcal{L}^5$ (up to $\log \mathcal{L}$). Section 10 shows that the junta term cannot be reduced within the present architecture.

10. THE CEILING OF THE MINORANT ARCHITECTURE

The proof of Theorem 1.1 passes through a minorant $B \leq F$ of positive Haar mean whose cells have modulus $e^{O(\mathcal{LS})}$. Here we show that no minorant of smaller level exists, dense or sparse, on any fibre.

Setting. Fix a modulus Q and a unit r modulo Q , and consider the fibre $\{n \in \hat{\mathbb{Z}}^\times : n \equiv r \pmod{Q}\}$ with its Haar probability. Let $F(n) = \mathbf{1}[W(n) > T]$. A function on the fibre has *level* at most D if it is a linear combination of functions $n \mapsto \phi(n \bmod qQ)$ with $q \leq D$. Every cell of modulus at most D and every character of conductor at most D (restricted to the fibre) has level at most D ; the minorant of Theorem 9.1 has level at most $\max_i d_i \leq T^3 e^{2\tau}$. All functions below depend on n modulo a fixed finite modulus, so all probability spaces are finite.

Lemma 10.1 (Planting; proved). *Let $b_1, \dots, b_N$ be independent bits with $\mathbb{P}(b_i = 1) = p_i \in [0, 1)$, and law ν_0 . Put $r_i = p_i/(1 - p_i)$, $R = \sum_i r_i$ and $r^* = \max_i r_i$. Let $k \geq 0$ with*

$$R \geq (k + 1) + (2k + 1)r^*. \quad (11)$$

Then there is a probability law ν on $\{0, 1\}^N$ with the same marginals as ν_0 on every set of at most k coordinates and $\nu(b = 0) = 0$.

Proof. Write 1_y for the configuration with ones exactly on $y \subseteq [N]$, and $P_0 = \nu_0(1_\emptyset) = \prod(1 - p_i)$, so $\nu_0(1_y) = P_0 \prod_{i \in y} r_i$. Let e_a denote the elementary symmetric functions. If $e_{k+1}(r) = 0$, at most k of the r_i are positive and $R \leq kr^*$, contradicting (11). For $|J| = k + 1$ put $w_J = \prod_{i \in J} r_i / e_{k+1}(r)$, so $\sum_J w_J = 1$, and

$$\nu = \nu_0 + P_0 \sum_{|J|=k+1} w_J \sigma_J, \quad \sigma_J = \sum_{y \subseteq J} (-1)^{|y|+1} \delta_{1_y}.$$

Marginals. For $|K| \leq k$ and $z \subseteq K$, the σ_J -mass of $\{x : x_K = 1_z\}$ is $[z \subseteq J] \sum_{y' \subseteq J \setminus K} (-1)^{|z|+|y'|+1} = 0$, as $J \setminus K \neq \emptyset$. In particular ($K = \emptyset$) ν has total mass 1. *Zero.* $\nu(1_\emptyset) = P_0 - P_0 \sum_J w_J = 0$. *Positivity.* Only $y \subseteq J$ with $|y| = j \geq 2$ even receives negative charge, of total $P_0 \sum_{J \supseteq y} w_J = P_0 \prod_{i \in y} r_i e_{k+1-j}(s) / e_{k+1}(r)$, where $s = (r_i)_{i \notin y}$. So it suffices that $e_{k+1-j}(s) \leq e_{k+1}(r)$. Now $e_{k+1}(r) \geq e_{k+1}(s)$, and $e_a(s) \geq e_{a-1}(s)$ for $1 \leq a \leq k+1$: each a -set arises a times as $I \cup \{i\}$ with $|I| = a-1$, $i \notin I$, so $a e_a(s) \geq e_{a-1}(s)(\sum s - (a-1)r^*)$, and $\sum s - (a-1)r^* \geq R - (j+a-1)r^* \geq R - (2k+1)r^* \geq k+1 \geq a$. Chain $a = k+2-j, \dots, k+1$. $\square$

This is the dual (linear programming) side of lower-bound sieves: minorants of degree k of $\mathbf{1}[b = 0]$ with positive mean exist only if no such ν exists (the duality itself is standard, cf. [38, Thm. 5]). It is close in spirit to work on k -wise independent laws with extremal probabilities [4, 34]; we claim no novelty for it. In the Poisson-like model with R of order S , Bonferroni inequalities of order $\asymp S$ do give positive minorants, so the order $k \asymp R$ is right.

Lemma 10.2 (Level barrier; proved). *Let independent coordinates be split into small ones X_s and big ones $(X_b)_{b \in \mathfrak{B}}$. Let $\mathcal{F}$ be a family of events, each of the form $E = \{X_s \in \Sigma_E\} \cap \{X_{b(E)} \in \Gamma_E\}$ with exactly one big coordinate $b(E)$, and let F be the indicator that no event of a family $\mathcal{E} \supseteq \mathcal{F}$ occurs. For a value x_s put $\Omega_b(x_s) = \bigcup_{E: b(E)=b, x_s \in \Sigma_E} \Gamma_E$, $p_b(x_s) = \mathbb{P}(X_b \in \Omega_b(x_s))$, $r_b = p_b/(1 - p_b)$ and $R(x_s) = \sum_b r_b(x_s)$. Suppose $p_b(x_s) \leq p^* < 1$ for all b, x_s , put $r^* = p^*/(1 - p^*)$, and suppose $R(x_s) \geq (k+1) + (2k+1)r^*$ for every x_s . Let B be a linear combination of functions each depending on X_s and on at most k big coordinates. If $B \leq F$ pointwise, then $\mathbb{E}B \leq 0$.*

Proof. Define a law ν : draw x_s from its true law; draw bits $(b_b)_b$ from the law of Lemma 10.1 for the independent bits $\mathbf{1}[X_b \in \Omega_b(x_s)]$ (bits with $p_b = 0$ have $w_J = 0$ and are never set); given the bits, draw each X_b independently from its true law conditioned on $X_b \in \Omega_b(x_s)$ (bit 1) or $X_b \notin \Omega_b(x_s)$ (bit 0). For a function ϕ of x_s and of the big coordinates in a set K with $|K| \leq k$: given x_s , the bits on K are independent with the true marginals, so $(X_b)_{b \in K}$ has its true product law, and $\mathbb{E}_\nu \phi = \mathbb{E} \phi$. Hence $\mathbb{E}B = \mathbb{E}_\nu B \leq \mathbb{E}_\nu F$. Under ν some bit is 1, so some event of $\mathcal{F}$ occurs and $F = 0$. $\square$

The ES instance. Fix a small absolute $\varepsilon \in (0, 1/20]$ (chosen in Lemma 10.4), and put $X = T^\varepsilon$, $V = X^{1/3}$, $L = \log X = \varepsilon \mathcal{L}$ and $y = \mathcal{L}^6$. With D^* as in §4, let $\mathcal{F}^*$ be the set of events $E_{v\ell, D}$ with

$$n_D := D^* \in [V, X], \quad v \leq V \text{ squarefree and } y\text{-rough}, \\ \ell \in (T^{0.6}, T^{0.7}] \text{ prime}, \quad v\ell \equiv -1 \pmod{4n_D}.$$

The congruence gives $v\ell \equiv 3 \pmod{4}$ and $n_D \mid A_{v\ell}$, i.e. $D \mid A_{v\ell}^2$; and $D \leq n_D^2 \leq T^{2\varepsilon}$. On a fibre $n \equiv r \pmod{Q}$ the *big* coordinates are the $X_\ell = n \bmod \ell$ for primes $\ell > T^{0.6}$ with $\ell \nmid Q$; everything else is small (coordinates fixed by the fibre are constants). A modulus $q \leq D$ has at most $\log D/(0.6\mathcal{L})$ prime factors $> T^{0.6}$, so a function of level at most D is a combination of functions of the small coordinates and of at most $k = \lfloor \log D/(0.6\mathcal{L}) \rfloor$ big ones.

Lemma 10.3 (Unique small part; proved). *For each pair (ℓ, D) there is at most one v with $E_{v\ell, D} \in \mathcal{F}^*$. Every event of $\mathcal{F}^*$ has exactly one big coordinate (if $\ell \nmid Q$), $p_\ell(x) \leq p^* := T^{-1/2}$ for all ℓ and x , and, for every small configuration x ,*

$$\sum_{\ell > T^{0.6}} p_\ell(x) = \sum_v \sum_{\substack{D: n_D \in [V, X] \\ 4D \equiv -x \pmod{v}}} c(v, n_D), \quad c(v, n) = \sum_{\substack{\ell \in (T^{0.6}, T^{0.7}] \\ \ell \equiv -v^{-1} \pmod{4n}}} \frac{1}{\ell - 1},$$

with v over the y -rough squarefree $v \leq V$, $c(v, n) = 0$ if $\gcd(v, 2n) > 1$, and $p_\ell(x)$ computed with X_ℓ uniform on the units modulo ℓ .

Proof. $v \equiv -\ell^{-1} \pmod{4n_D}$ and $1 \leq v \leq V \leq n_D < 4n_D$, so v is determined by (ℓ, n_D) . As $v < T^{0.6} < \ell$, ℓ is the only prime factor of $v\ell$ above $T^{0.6}$.

Given x , the events at ℓ whose small part holds are the $E_{v\ell,D}$ with $x \equiv -4D \pmod{v}$, and distinct D give distinct classes $-4D \pmod{\ell}$, as $0 < |D - D'| < T^{2\varepsilon} < \ell$. So $p_\ell(x) = (\ell - 1)^{-1} \#\{D : v(\ell, D) \text{ exists, } x \equiv -4D \pmod{v}\}$; this is at most $X(1 + L)/(\ell - 1) \leq T^{-1/2}$ by (F3). Sum over ℓ and group by v . $\square$

Lemma 10.4 (Class-uniform primes; proved modulo Theorems 5.1 and 5.2). *There is an absolute $\varepsilon_0 > 0$ such that for $\varepsilon \leq \varepsilon_0$ and T large the following holds. Apply Theorem 5.2 with $Q_G = 4X$, let q_1 be the conductor of the exceptional character if there is one, and put $\mathcal{N}_{\text{exc}} = \{n : q_1 \mid 4n\}$ if q_1 exists and $q_1 > \mathcal{L}^{1.9}$, and $\mathcal{N}_{\text{exc}} = \emptyset$ otherwise. Then $c(v, n) \geq 0.13/\varphi(4n)$ for every $n \in [V, X] \setminus \mathcal{N}_{\text{exc}}$ and every $v \leq V$ with $\gcd(v, 2n) = 1$.*

Proof. Put $q = 4n \leq Q_G$ and $b \equiv -v^{-1} \pmod{q}$. Each $\chi \pmod{q}$ is induced by a primitive χ^* , distinct χ give distinct χ^* , and $|\vartheta(x; \chi) - \vartheta(x; \chi^*)| \leq \log q$. By orthogonality and Theorem 5.2, for $x \in [T^{0.6}, T^{0.7}]$,

$$\left| \vartheta(x; q, b) - \frac{x}{\varphi(q)} + [q_1 \mid q] \chi_1(b) \frac{x^{\beta_1}}{\beta_1 \varphi(q)} \right| \leq \frac{\text{Err}_G(x)}{\varphi(q)} + \log q,$$

where $\text{Err}_G(x)$ is the right-hand side of Theorem 5.2 (times C_G), and the bracket is present only in the exceptional case. The range condition $Q_G^{6c} \leq x$ holds for $\varepsilon \leq 0.05/c$, and $\log x / \log Q_G \geq 0.6/(\varepsilon + o(1))$. Without an exceptional zero, $\text{Err}_G(x) \leq C_G x(e^{-0.6/(\kappa(\varepsilon + o(1)))} + o(1))$. With one, $(1 - \beta_1) \log x \leq \log x / (\kappa \log Q_G) \leq 0.7/(\kappa(\varepsilon + o(1)))$, so $\text{Err}_G(x) \leq C_G x(0.7/(\kappa\varepsilon))(e^{-0.6/(\varepsilon + o(1))} + o(1))$. Both are at most $x/16$ for ε small, and $\log q = o(x/\varphi(q))$. If $q_1 \mid q$ and $q_1 \leq \mathcal{L}^{1.9}$, Theorem 5.1 gives $1 - \beta_1 \gg \mathcal{L}^{-0.95}(\log \mathcal{L})^{-2}$, so $x^{\beta_1 - 1} = o(1)$; if $q_1 \mid q$ and $q_1 > \mathcal{L}^{1.9}$, then $n \in \mathcal{N}_{\text{exc}}$. So $\vartheta(t; q, b) \in [\frac{7}{8} - o(1), \frac{9}{8} + o(1)] t/\varphi(q)$ on $[T^{0.6}, T^{0.7}]$, and partial summation against $1/(t \log t)$ gives $c(v, n) \geq \sum 1/\ell \geq \frac{7}{8} \log \frac{7}{6} / \varphi(q) - O(1/(\mathcal{L}\varphi(q))) \geq 0.13/\varphi(q)$. $\square$

If $q_1 > \mathcal{L}^{1.9}$, put $q'' = q_1 / \gcd(q_1, 4) \geq q_1/4$; then $n \in \mathcal{N}_{\text{exc}}$ if and only if $q'' \mid n$, and by the divisor bound $\tau(q'')/q'' \leq q''^{-0.95} \leq 4\mathcal{L}^{-1.8}$ for T large.

Lemma 10.5 (Class-uniform D ; proved). *For T large, every odd squarefree y -rough $v \leq V$ and every unit a modulo v ,*

$$W(v, a) := \sum_{\substack{D: n_D \in [V, X] \setminus \mathcal{N}_{\text{exc}} \\ D \equiv a \pmod{v}}} \frac{1}{\varphi(4n_D)} \geq \frac{L^2}{200v}.$$

Proof. Write $D = \kappa t^2$ with κ squarefree, so $n_D = \kappa t$. Keep only $t \leq X^{1/6}$ with $(t, v) = 1$ and $\kappa \in (K_1, K_2]$, $K_1 = X^{2/3}$, $K_2 = X/t$; then $n_D \in [V, X]$, and $D \equiv a \pmod{v}$ means $\kappa \equiv b := at^{-2} \pmod{v}$. As $\varphi(4n) \leq 2n$, $W(v, a) \geq H_{\text{all}} - H_{\text{exc}}$, where H_{all} and H_{exc} are the sums of $1/(2\kappa t)$ over the kept pairs, with all n and with $n \in \mathcal{N}_{\text{exc}}$ respectively. For a unit b modulo v , the number of squarefree $\kappa \leq K$ with $\kappa \equiv b \pmod{v}$ is $\sigma_v K/v + O(\sqrt{K})$ with $\sigma_v = \sum_{(d,v)=1} \mu(d)/d^2 \geq 6/\pi^2$ (write $\mu^2(\kappa) = \sum_{d^2 \mid \kappa} \mu(d)$). Partial summation gives $\sum_{\kappa \in (K_1, K_2], \kappa \equiv b} 1/\kappa \geq (\sigma_v/v) \log(X^{1/3}/t) - O(K_1^{-1/2})$. For y -rough

$v, \sum_{t \leq X^{1/6}, (t,v)=1} t^{-1} \log(X^{1/3}/t) \geq L^2/24 - O(\omega(v)L^2/y) - O(L) \geq L^2/48$. So $H_{\text{all}} \geq \frac{1}{2} \cdot \frac{6}{\pi^2} \cdot \frac{L^2}{48v} - O(LX^{-1/3})$. If $\mathcal{N}_{\text{exc}} \neq \emptyset$: $q'' \mid \kappa t$ forces $q''/(q'', t) \mid \kappa$, which with $\kappa \equiv b \pmod{v}$ is one class modulo $vq''/(q'', t)$ or empty, so by (F4) the $1/\kappa$ -sum is at most $1/K_1 + (q'', t)L/(vq'')$. Since $\sum_{t \leq X} (q'', t)/t \leq \tau(q'')(1+L)$, we get $H_{\text{exc}} \leq L^2\tau(q'')/(vq'') + O(LX^{-2/3}) \leq 4L^2\mathcal{L}^{-1.8}/v$. As $X^{-1/3} \leq 1/v$, $W(v, a) \geq (L^2/v)(6/(96\pi^2) - o(1)) \geq L^2/(200v)$. $\square$

Theorem 10.6 (No positive low-level minorant; proved modulo Theorems 5.1, 5.2 and the fundamental lemma). *There are absolute constants $c, \varepsilon > 0$ such that for T large, every modulus Q with $\log Q \leq T^{0.05}$ and every unit r modulo Q the following holds. If B is a function on the fibre $n \equiv r \pmod{Q}$ of level at most D , with $B \leq F$ pointwise and $\log D \leq c\mathcal{L}^4/\log \mathcal{L}$, then $\mathbb{E}B \leq 0$.*

Proof. Use $\mathcal{F}^*$ without the events whose ℓ divides Q ; each of its events has exactly one big coordinate, and $p^* \leq T^{-1/2}$ (Lemma 10.3). Fix a small configuration x ; its entries are units. By Lemma 10.3, $R(x) \geq \sum_{\ell} p_{\ell}(x) \geq \sum_v \sum_{D \equiv -x/4 \pmod{v}, n_D \notin \mathcal{N}_{\text{exc}}} c(v, n_D) - \omega_{\text{big}}(Q)p^*$, where $\omega_{\text{big}}(Q) \leq \log Q/(0.6\mathcal{L}) \leq T^{0.05}$ counts the prime factors of Q above $T^{0.6}$. For $D \equiv -x/4 \pmod{v}$, D is a unit modulo v , so $\gcd(v, 2n_D) = 1$, and Lemmas 10.4 and 10.5 give

$$R(x) \geq 0.13 \cdot \frac{L^2}{200} \sum_{\substack{v \leq V \\ y\text{-rough, squarefree}}} \frac{1}{v} - T^{0.05}T^{-1/2} \geq \mu^* := c_9 \varepsilon^3 \frac{\mathcal{L}^3}{\log \mathcal{L}},$$

where the v -sum is $\geq c_3 \log V/\log y$ by the fundamental lemma (as in Lemma 4.5; squarefreeness costs $O(\mathcal{L}/y)$). With $k = \lfloor \log D/(0.6\mathcal{L}) \rfloor$ we have $(k+1) + (2k+1)r^* \leq (k+1)(1+4p^*) \leq \mu^*$ if $c \leq 0.3c_9\varepsilon^3$. Lemma 10.2, applied with $\mathcal{E}$ the set of all events with $M \leq T$, gives $\mathbb{E}B \leq 0$. $\square$

The mechanism: conditioned on the coordinates below $T^{0.6}$, what remains is a sieve on the big primes with removed classes of total density $\gg \mathcal{L}^3/\log \mathcal{L}$ (the bulk of the ES mass comes from moduli with one large prime factor), so a sieve of dimension $\kappa \asymp \mathcal{L}^3/\log \mathcal{L}$ with all moduli above $T^{0.6}$. Lower-bound sieves of dimension κ need level about z^κ , here $e^{\asymp \mathcal{L}^4/\log \mathcal{L}}$. Lemma 10.1 is the elementary form of this, and the uniqueness family makes $R(x)$ large for every small configuration, not only on average.

Corollary 10.7 (The $1/4$ ceiling; proved implication). *Consider any argument that produces a hard prime p with $W(p) > T$ from a function $B \leq F$ on a fibre $n \equiv r \pmod{Q}$, of level at most Z and with $\mathbb{E}B > 0$, through a transfer theorem whose hypothesis requires $\log x \geq C \log Z$ for the search bound x (with $Z \geq Q$). Every such argument has $\log x \gg \mathcal{L}^4/\log \mathcal{L}$. Theorem 6.1, and hence the proof of Theorem 1.1, is of this type, whatever the minorant (sandwich, Selberg, Bonferroni, dense or sparse, any value of A). So such arguments cannot give $W(p) \geq \exp((\log p)^{1/4+\delta})$ for any $\delta > 0$, and Theorem 1.1 is optimal among them up to a factor $(\log \log p)^{1/2}$ in $\log W(p)$.*

Proof. If $\log Q > T^{0.05}$, then $\log x \geq C \log Z \geq C \log Q > e^{0.05\mathcal{L}}$. Otherwise Theorem 10.6 forces $\log Z > c\mathcal{L}^4/\log \mathcal{L}$. In both cases $\log x \gg \mathcal{L}^4/\log \mathcal{L}$, so the certified statement “some hard $p \leq x$ has $W(p) > T$ ” holds at best with $\mathcal{L} \ll (\log x \log \log x)^{1/4}$. $\square$

Scope. Corollary 10.7 says nothing about the true size of the least such prime; nothing about prime input other than low-conductor minorants (bilinear or Type II information, parity-sensitive input, or mixtures of majorants and minorants); nothing about transfers whose positivity comes from an exceptional-character term rather than from $\mathbb{E}B > 0$ (such transfers can only work at Siegel scales); and nothing about functions that are not of bounded level in the above sense.

Corollary 10.8 (The energy tail cannot be improved; proved modulo Theorems 5.1, 5.2, 3.1 and the fundamental lemma). *On the fibre of Theorem 3.3, there is no bound $\mathbb{E}(F^{(j)} - u_j)^2 \leq 2^{-\tau/\rho}$ for all j and all τ , with u_j as in (10) and $\rho \leq c'\mathcal{L}/(\log \mathcal{L})^2$ (c' small absolute). So the weights of Theorem 7.9 are optimal for this purpose up to a factor $(\log \mathcal{L})^2$.*

Proof. Such a bound gives $\mathbb{E}[F - B] \leq m^2 S 2^{-\tau/\rho} \leq \delta/100$ for some $\tau \ll \rho(S_1 + \mathcal{L}) \ll \rho\mathcal{L}^3 \log \mathcal{L}$, so $\mathbb{E}B \geq 0.99\delta > 0$ (Lemma 8.6), and B has level at most $T^3 e^{2\tau}$ (Lemma 8.4). For c' small this contradicts Theorem 10.6. $\square$

Remark 10.9 (The heuristic exponent 1/3; Assessment). By Theorems 3.3 and 4.4, $\log(1/\delta^*(T)) = \mathcal{L}^{3+o(1)}$. If the least prime in the avoiding set behaved like the least prime in a set of Haar density δ^* (as for a single progression, by Linnik’s theorem), one would expect a hard prime with $W(p) > T$ and $\log p \ll \mathcal{L}^{3+o(1)}$, i.e. $\log W(p) \approx (\log p)^{1/3}$ infinitely often. This is a heuristic, not a claim. By Corollary 10.7 it cannot be reached by a level- Z minorant with a transfer costing $\log x \gg \log Z$: one would need prime-detecting input for a sifted set of sieve dimension $\asymp \mathcal{L}^3$ at level well below its sieving limit, that is, information beyond minorants of bounded level, such as bilinear (Type II) estimates or other parity-sensitive input adapted to the ES system. We do not know how to provide it.

11. SUMMARY OF STATUS

- Proved: Lemmas 2.1, 2.2, 2.3, 2.5, 2.8 (with the classical Lemma 2.7), 2.9, Lemma 3.2(B); the Janson-type inequality (Lemmas 4.1, 4.2, Theorem 4.3) and Lemmas 4.6, 4.7; the energy bound (Lemmas 7.2–7.4, Theorems 7.5 and 7.9, Corollaries 7.6 and 7.10, Remark 7.7, and Remark 7.8 except the cited switching-lemma constant); Lemmas 8.1, 8.2, 8.3, 8.4, 8.5, 8.6; the planting and level-barrier Lemmas 10.1, 10.2, and Lemmas 10.3, 10.5.
- Proved modulo the Nair–Tenenbaum theorem (Theorem 3.1): part (A) of Lemma 3.2 and the Haar upper bound Theorem 3.3.

- Proved modulo the fundamental lemma of sieve theory: Lemma 4.5 and the Haar lower bound Theorem 4.4. With the previous item: Theorem 1.2 (the Haar exponent is 3).
- Proved modulo Gallagher’s theorem (Theorems 5.2 and 5.1, quoted from the draft [26]): the linear transfer Theorem 6.1 and Lemma 10.4.
- Proved modulo Gallagher’s theorem and Theorem 3.1: Theorem 9.1 and Theorem 1.1.
- Proved modulo Gallagher’s theorem and the fundamental lemma: Theorem 10.6 (= Theorem 1.3). Proved implications: Corollary 10.7 (within its stated scope) and Corollary 10.8 (which also uses Theorem 3.1).
- Evidence, not used: the computer check of Lemma 2.2 for $M \leq 2 \cdot 10^5$; the exact check of Theorem 4.3 on small instances; the Monte Carlo fit of $\delta^*(T)$ for $T \leq 32767$.
- Assessment: Remarks 6.2, 9.2 and 10.9.
- Not claimed: anything about the Erdős–Straus conjecture; the heuristic exponent $1/3$; anything about arguments outside the scope of Corollary 10.7; any numerical instance; priority for Theorem 7.5 or Theorem 4.3 (literature searches partial).
- The Elsholtz–Tao estimates, Håstad’s switching lemma and the Thorner–Zaman theorem are no longer used; they are cited only for comparison.

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